

Pound-Drever-Hall Locking: A Frequency-Domain Model

PDH learning project — *source/PDH*

September 7, 2026

Abstract

This is a slow, worked-example companion to the MATLAB model built under *source/PDH*, whose purpose is to *learn* Pound-Drever-Hall (PDH) laser frequency stabilization by reproducing the physics of the Liquid Instruments webinar “High-Performance PDH Locking with Reconfigurable Instrumentation” (J. Miller, 2024) section by section, in code, and cross-checking every derivation numerically rather than trusting it by inspection. Section 1 covers the passive Fabry-Pérot cavity: where its three characteristic numbers (FSR, Finesse, linewidth) come from, and where the reflection/transmission formulas themselves come from, derived as a geometric series over round trips. Section 2 covers phase modulation: why PDH needs sidebands at all, the Jacobi-Anger expansion that produces them, and the phasor picture that makes the counter-rotating sidebands visualizable. Section 3 builds the explicit photodiode/mixer/low-pass-filter detection chain that turns the reflected field into the PDH error signal. Section 4 reframes that same error signal as a single geometric dot product between the carrier’s own reflected phasor and the sideband resultant, worked through at four characteristic detunings. This completes the frequency-domain model the underlying wayfinder map (*.scratch/pdh-frequency-domain-model/*) set out to build; the closed-loop time-domain servo simulation (PI controller, simulated PZT actuator, lock acquisition) is a separate, later effort.

1 The Fabry-Pérot Cavity

1.1 The physical picture

A Fabry-Pérot cavity is two mirrors facing each other, separated by a length. Send a laser in through one mirror. The light bounces back and forth: a little bit leaks out through each mirror on every pass, and what leaks out interferes with what leaked out on previous passes. Whether that interference is constructive or destructive depends entirely on how many wavelengths fit exactly into one round trip — which depends on the laser’s frequency. That is the entire mechanism: frequency selects resonance because frequency sets the round-trip phase.

Everything about the cavity’s response reduces to three quantities, computed by the *OpticalCavity* class from four physical inputs: the cavity length, the laser wavelength, and the two mirrors’ power reflectivities. The reflection and transmission formulas derived below follow [1], Eqs. 5–6. Table 1 collects the notation.

Symbol	Code identifier	Meaning
L	<i>CavityLength</i>	mirror separation (round-trip length), m
λ	<i>Wavelength</i>	laser vacuum wavelength, m
$n(\lambda)$	<i>RefractiveIndex</i>	medium refractive index; scalar or dispersive function handle
R_1, R_2	<i>InputMirrorPowerReflectivity, BackMirrorPowerReflectivity</i>	mirror power reflectivities
r_1, r_2	(private) <i>mirrorAmplitudeReflectivities</i>	amplitude reflectivities, $r_i = \sqrt{R_i}$
t_1, t_2	—	amplitude transmissivities, $t_i = \sqrt{1 - R_i}$ (lossless mirror)
g	—	round-trip amplitude reflectivity, $g = r_1 r_2$
FSR	<i>get.FSR</i>	free spectral range, Hz (Eq. (1))
N	—	integer resonance mode number (Eq. (2)); <i>not</i> the round-trip index k of Eqs. (6)–(7)
f_{laser}	—	the laser’s actual frequency (Eq. (3))
$f_{0,N}$	—	the N -th resonance’s frequency (Eq. (2))
Δf_{detune}	argument <i>detuningHz</i>	laser detuning from the nearest cavity resonance, Hz (Eq. (3))
φ	<i>roundTripPhase</i>	round-trip phase accumulated at detuning Δf_{detune}
$\mathcal{F}$	<i>get.Finesse</i>	cavity finesse (Eq. (4))
$\Delta \nu_{\text{FWHM}}$	<i>get.LinewidthFWHM</i>	resonance linewidth, FWHM, Hz (Eq. (5))
$r(\varphi)$	<i>reflectionCoefficient</i>	the cavity’s overall complex reflection coefficient (Eq. (7)) — not to be confused with the per-mirror r_1, r_2 above
T_1, T_2	—	mirror power transmissivities, $T_i = 1 - R_i$ (§1.6)

Table 1: Notation used throughout Section 1.

1.2 The three numbers: FSR, Finesse, linewidth

Free Spectral Range (FSR). The frequency spacing between one resonance and the next. The round-trip time is $2n(\lambda)L/c$ (there and back, through a medium of index $n(\lambda)$), so the light returns “in phase” only at frequencies spaced by the round-trip rate:

$$\text{FSR} = \frac{c}{2n(\lambda)L}. \quad (1)$$

For the running example used throughout this report — $L = 14.989\,622\,9$ cm, $\lambda = 1064$ nm, vacuum ($n = 1$) — this evaluates to exactly $\text{FSR} = 1$ GHz.

The cavity’s resonances form a discrete comb of longitudinal modes, indexed by an integer mode number N , with resonance frequencies

$$f_{0,N} = f_{0,0} + N \cdot \text{FSR}, \quad (2)$$

spaced by exactly one FSR apart. The laser’s detuning from whichever mode it currently sits nearest is

$$\Delta f_{\text{detune}} = f_{\text{laser}} - f_{0,N}, \quad (3)$$

the same quantity passed as *detuningHz* throughout *OpticalCavity* (Table 1).

Finesse. How sharp each resonance is, in units of “how many resonances would fit across one linewidth.” Better mirrors (higher reflectivity) keep light bouncing longer, so a resonance can be a narrower sliver of frequency before it stops building up:

$$\mathcal{F} = \frac{\pi\sqrt{g}}{1-g}, \quad g = r_1 r_2. \quad (4)$$

This is *get.Finesse*, line-for-line.

Linewidth (FWHM). The actual width, in Hz, of a resonance dip or peak:

$$\Delta\nu_{\text{FWHM}} = \frac{\text{FSR}}{\mathcal{F}}. \quad (5)$$

This is *get.LinewidthFWHM* — almost a tautology once Eqs. (1)–(4) exist, since Finesse is *defined* as that ratio.

Table 2 makes the Finesse $\leftrightarrow$ reflectivity $\leftrightarrow$ linewidth relationship concrete: FSR is held fixed at 1 GHz while the (matched) mirror reflectivity improves. Each $10\times$ improvement in mirror quality ($1 - R$ shrinking by $10\times$) gives roughly $10\times$ the Finesse and $10\times$ narrower linewidth. The last column, $\pi/(1 - R)$, is the textbook high-reflectivity approximation to Eq. (4); it only approaches the exact value as $R \rightarrow 1$, since it drops the $\sqrt{g} \rightarrow 1$ approximation’s own residual error.

$R = R_1 = R_2$	Finesse	Linewidth (FWHM)	$\pi/(1 - R)$
0.9000	29.8	3.355×10^7 Hz	31.4
0.9900	312.6	3.199×10^6 Hz	314.2
0.9990	3140.0	3.185×10^5 Hz	3141.6
0.9999	31414.4	3.183×10^4 Hz	31415.9

Table 2: Finesse and linewidth vs. matched mirror power reflectivity R , at fixed $\text{FSR} = 1$ GHz. Generated by *generateReportFigures.m* (*exportFinesseTable*) directly from *OpticalCavity*, so it can never silently drift from the code.

1.3 Deriving the reflection and transmission coefficients

This is the part worth understanding, not just trusting. Inside the cavity, light does not bounce just once — it bounces over and over, and every one of those escaping “echoes” adds up coherently. Each round trip multiplies the intracavity field by $g e^{i\varphi}$ (mirror reflectivities times the phase $\varphi = 2\pi\Delta f_{\text{detune}}/\text{FSR}$ picked up traversing the round-trip length at detuning Δf_{detune}). So the transmitted field is a geometric series over round-trip number k (deliberately not N , reserved above for the resonance mode number, Eq. (2)):

$$\frac{E_{\text{trans}}}{E_{\text{in}}} = t_1 t_2 \sum_{k=0}^{\infty} (g e^{i\varphi})^k = \frac{t_1 t_2}{1 - g e^{i\varphi}}. \quad (6)$$

This is exactly *transmissionCoefficient*’s formula. Figure 1 does not take this on faith: it sums the series numerically to a finite number of round trips and plots the residual error against the exact closed form, on resonance, for two different finesse values. The convergence rate is set by how quickly g^k decays — controlled by the same $1 - g$ that sets the Finesse in Eq. (4). High finesse literally means “light survives many round trips before it decays,” and that is why the resonance is narrow: a signal that persists longer in time is confined to a narrower band in frequency, the same time $\leftrightarrow$ frequency tradeoff behind every resonance.

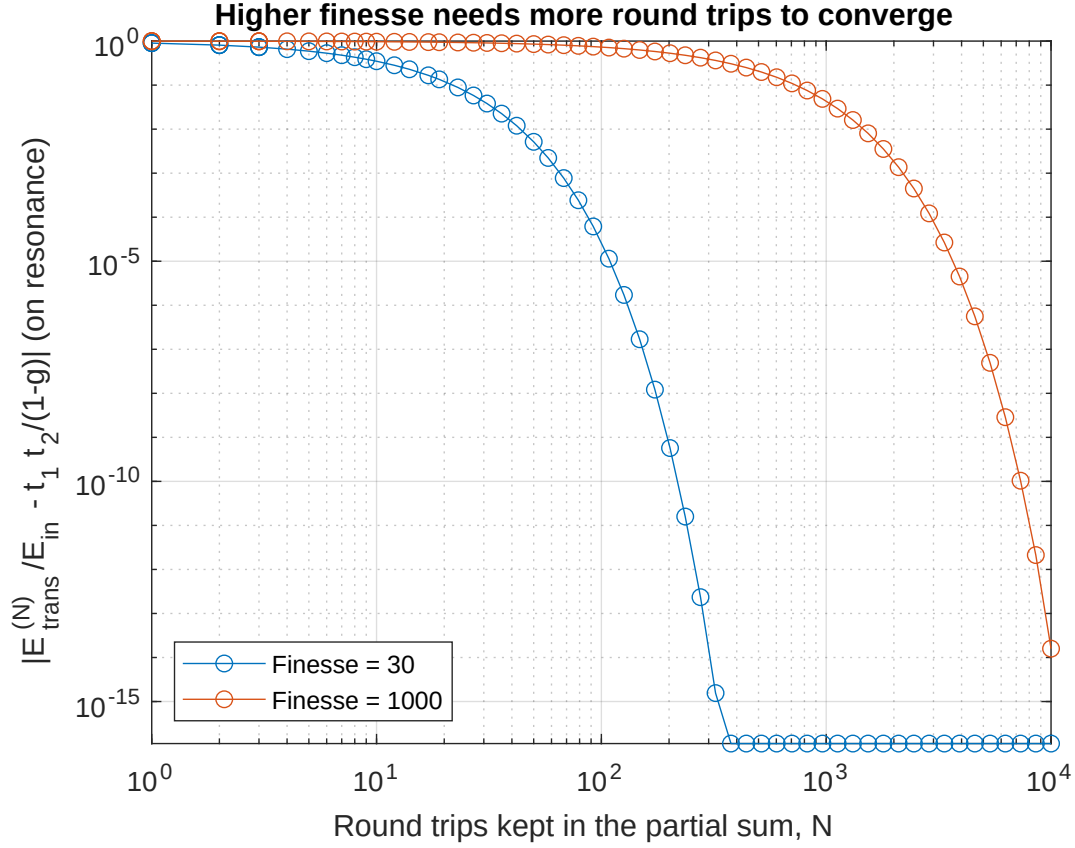


Figure 1: Convergence of the truncated geometric series (Eq. (6), first k terms) to the exact closed form, on resonance, for a low-finesse and a high-finesse cavity. The high-finesse cavity needs roughly 100 $\times$ more round trips to reach the same residual error — the numerical face of “narrower linewidth needs more round trips to build up.”

Reflection: one extra term, same idea. The transmitted field above is only the leaked-out geometric series. The reflected field has that too, but it also gets the light that bounces straight off the front mirror without ever entering the cavity:

$$\frac{E_{\text{refl}}}{E_{\text{in}}} = -r_1 + t_1^2 r_2 e^{i\varphi} \sum_{k=0}^{\infty} (g e^{i\varphi})^k = \frac{r_2 e^{i\varphi} - r_1}{1 - g e^{i\varphi}} \quad (7)$$

after substituting $t_1^2 = 1 - r_1^2$ and simplifying. This is exactly *reflectionCoefficient*’s formula. For a symmetric cavity ($r_1 = r_2 = r$), setting $\varphi = 0$ gives $(r - r)/(1 - r^2) = 0$: the direct reflection off the front mirror exactly cancels the light leaking back out, on resonance. All the power that came in has to go somewhere, and with nothing reflected, it all goes through — exactly the “reflection dips to 0, transmission peaks to 1” picture in Figure 2 below. Note the structural difference from Eq. (6): the reflection formula has a $-r_1$ term sitting outside the round-trip sum; transmission is only the leaked-out buildup, reflection is that buildup *minus* the direct bounce.

1.4 Reading the cavity response

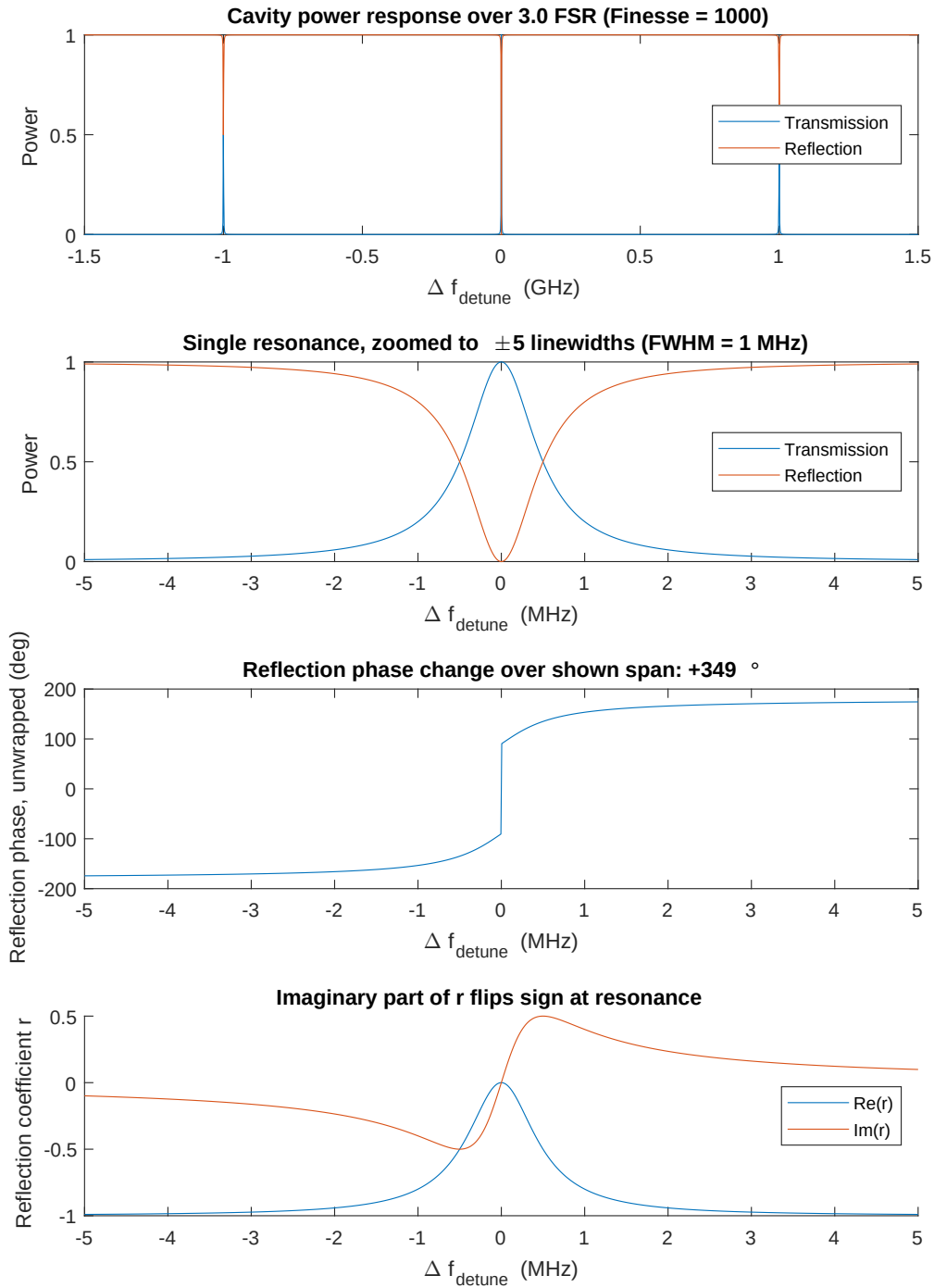


Figure 2: The running-example cavity's response (FSR = 1 GHz, $\mathcal{F} = 1000$), produced by *plotCavityResponse.m*.

Reading each panel of Figure 2 with the physics of Sections 1.2–1.3 in hand:

- **Panel 1** (wide scan, ± 1.5 FSR): three narrow spikes, one full GHz apart — that spacing is the FSR of Eq. (1). At this zoom the resonances look like infinitely thin lines because the linewidth (1 MHz) is a thousandth of the FSR (1 GHz) — a direct visual of “Finesse = 1000”: a thousand resonance-widths would fit between one resonance and the next.
- **Panel 2** (zoomed to ± 5 linewidths): the geometric series made visible. Transmission peaks exactly at 1 on resonance — light built up over thousands of round trips, all constructively interfering, escaping fully. One linewidth away, the round-trip phase φ has drifted enough that after many round trips the interference is no longer constructive: transmission collapses, reflection recovers. The width of that collapse is the linewidth of Eq. (5).
- **Panel 3** (reflection phase): the panel that matters most for PDH. Off resonance, the reflected light is basically just the direct bounce off the front mirror — nothing subtle. But near resonance, the reflected field’s *amplitude* passes through zero (Panel 2’s red curve), and amplitude passing through zero is a phase singularity: the phasor has to spin around to get from “pointing one way” to “pointing the opposite way” as detuning crosses zero. That spin is the steep 360° jump seen here. This phase response is the entire reason PDH works: it is a fast, sign-changing signal that can be extracted electronically and fed back to keep a laser locked to resonance — but a photodiode only measures power, not phase, which is exactly why Section 2 exists.
- **Panel 4** (real/imaginary parts of the reflection coefficient): the same phase information as Panel 3, decomposed into Cartesian components. $\text{Im}(r)$ is the antisymmetric, dispersive-shaped curve that flips sign exactly at resonance — this is the curve PDH’s error signal is built to reproduce electronically (Section 2 and later sections), and its shape is the direct payoff of Eq. (7) passing through zero.

1.5 A worked conceptual check: what destructively interferes with what

It is tempting to describe on-resonance cancellation as “all the round trips destructively interfere with each other.” That is not quite right, and the distinction matters for what follows.

Recall detuning Δf_{detune} (Eq. (3)), measured from whichever resonance mode (Eq. (2)) the laser currently sits nearest to — the three peaks in Figure 2’s Panel 1 are three such modes. It is what labels every panel of Figure 3 below.

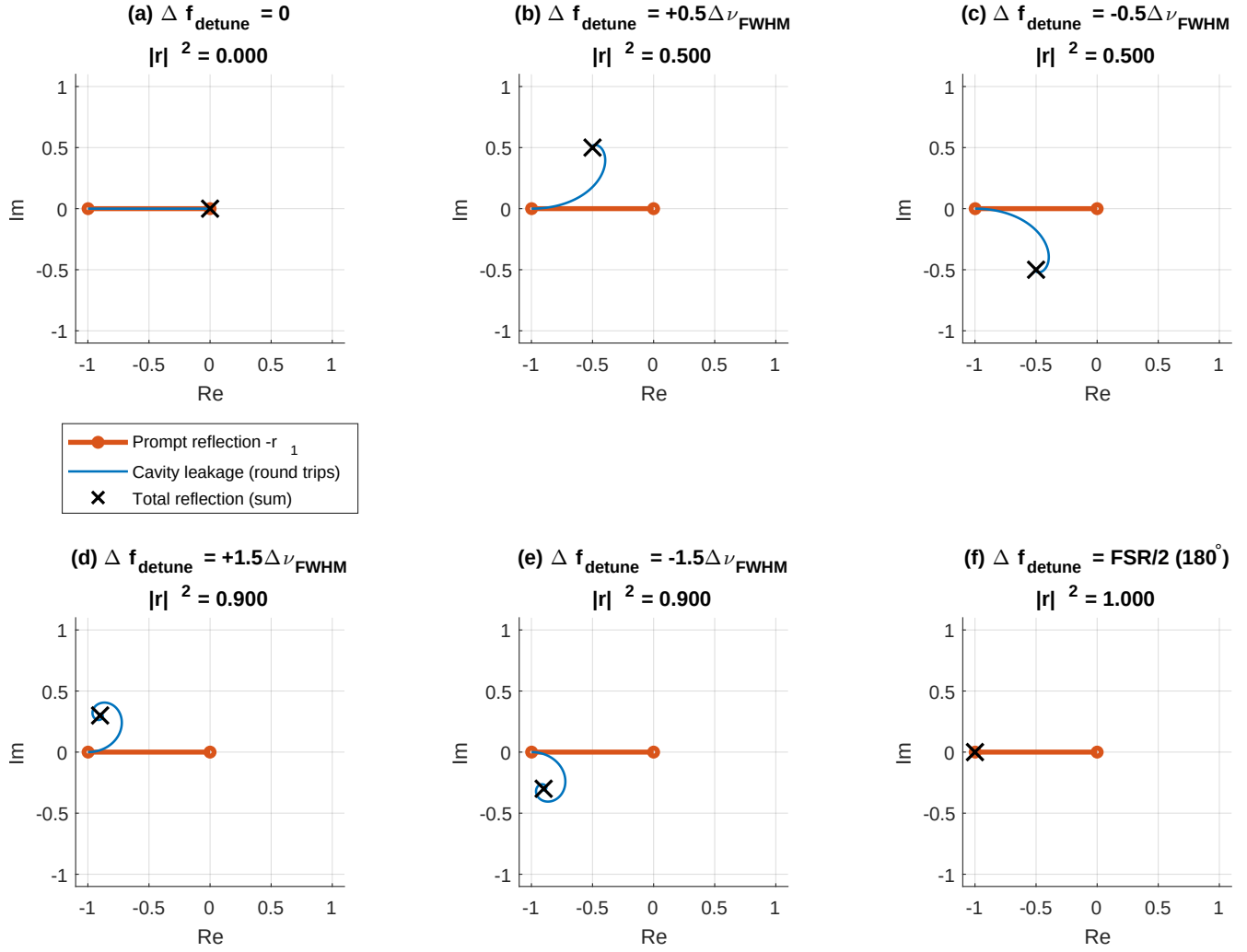


Figure 3: The round-trip phasor chain of Eq. (7), drawn head-to-tail, for this report’s running-example cavity ($\text{FSR} = 1 \text{ GHz}$, $\mathcal{F} = 1000$, the same cavity as Figure 2), at six fixed values of Δf_{detune} (Eq. (3)). Orange: the prompt reflection $-r_1$, the direct bounce off the front mirror, with filled endpoints at the origin and at $-r_1$. Blue: the cavity’s leaked-out buildup, the geometric series of Eq. (6)-like round-trip echoes — drawn as a smooth curve rather than a hand-countable chain of arrows, since at this finesse thousands of round trips are needed to converge (recall the Finesse= 1000 curve of Figure 1). Black x: the exact closed-form total, Eq. (7), confirming the sum lands in the right place. Panel (a) is on resonance ($\Delta f_{\text{detune}} = 0$); (b)/(c) are ± 0.5 and (d)/(e) are ± 1.5 of this cavity’s own linewidth (Eq. (5)) away from it; (f) is $\Delta f_{\text{detune}} = \text{FSR}/2$, the 180° anti-resonance point (maximum reflection, always exactly half an FSR away regardless of finesse). Detunings are expressed as fractions of the linewidth rather than fixed absolute degrees, since at $\mathcal{F} = 1000$ the resonance itself is only $360^\circ/1000 = 0.36^\circ$ of round-trip phase wide — fixed angles like $\pm 30^\circ$ or $\pm 90^\circ$ would all land far outside it and look identical (reflected power > 0.9999 in every case). As Δf_{detune} is swept continuously rather than sampled at six fixed points, the black x traces out a full circle in the complex plane — the *resonance circle* formalized in Section 1.6 (Figure 4).

Figure 3 makes the two-term structure of Eq. (7) literal. In every panel, the blue leakage curve itself always

builds *outward* from $-r_1$ — the individual round trips are interfering *constructively* with each other, building up coherently; if they destructively cancelled among themselves instead, nothing would build up and there would be no resonance to speak of. What changes from panel to panel is only where that curve’s final tip lands relative to the orange prompt-reflection arrow. In panel (a) (on resonance), the curve’s tip lands exactly opposite the tip of the orange arrow, as seen from the origin: added together, they cancel completely, $|r|^2 = 0$. That is a one-on-one cancellation between two vectors, not a many-way cancellation among the blue curve’s own contributions. Moving off resonance (panels (b)/(c), half a linewidth either side), the curve’s final tip rotates away from that exactly-opposite point, so the two vectors no longer fully cancel and $|r|^2$ rises to 0.5 — symmetric in sign, since ± 0.5 linewidths give the same $|r|^2$. Panels (d)/(e) (± 1.5 linewidths) show the curve rotated further still, well past the point of cancellation, $|r|^2 = 0.9$. Panel (f) (180° , anti-resonance) shows the curve’s tip swinging around to point in the *same* direction as the orange arrow — here the two vectors add rather than cancel, giving this cavity’s maximum reflection, $|r|^2 \approx 1$ (Section 1.2’s power conservation applies at every detuning, not only on resonance — at this much higher finesse than the earlier numeric example, almost none of the power is transmitted at the point of maximum reflection).

Detuning does not “destroy the reflection”; it destroys the *cancellation*, letting reflection recover back up toward 1 (numerically: at a fixed 1 MHz detuning, reflected power goes from 0.0035 at $\mathcal{F} = 30$ to 0.9997 at $\mathcal{F} \approx 31000$ — a higher-finesse cavity has a narrower resonance, so the same fixed detuning sits further outside it, and the interference there looks less and less destructive). At exact resonance, reflection is already at its minimum; detuning is what undoes the destructive interference and gives the reflected light back.

1.6 Over- and undercoupled cavities

Every cavity used so far has had matched mirrors, $R_1 = R_2$. That is a special case, called *critical coupling*, and real cavities do not have to satisfy it. Whether a cavity is under-, critically, or over-coupled compares two rates: how fast power couples in and back out through the front mirror, set by $T_1 = 1 - R_1$, against how fast power escapes by every other route — in this lossless two-mirror model, that means solely through the back mirror, set by $T_2 = 1 - R_2$. In terms of the mirror reflectivities already in use throughout this report:

- **Critically coupled** ($R_1 = R_2$): the two rates match. Setting $\varphi = 0$ in Eq. (7) gives $r(0) = 0$ exactly — every watt that enters the cavity eventually leaves through the back, none reflects. This is the cavity used throughout Sections 1.1–1.5.
- **Undercoupled** ($R_1 > R_2$): the front mirror reflects more than it lets through, relative to how much escapes at the back. On resonance $r(0) \neq 0$ — some input power simply reflects immediately, never fully absorbed into the buildup.
- **Overcoupled** ($R_1 < R_2$): the front mirror lets through more than the back can drain away. On resonance $r(0) \neq 0$ here too, but, as Figure 4 and Table 3 show, with the opposite sign relative to the far-off-resonance background $-r_1$.

Every example cavity in this report – including every one used above in Sections 1.1–1.5 – is one of six named fixtures on the *ExampleCavities* class: the three regimes above, each at a Low-Finesse ($\mathcal{F} \approx 30$) and a High-Finesse ($\mathcal{F} = 1000$ exactly) tier. Within a tier, the undercoupled and overcoupled fixtures are constructed to have *exactly* the critically coupled fixture’s Finesse: their mirror transmissivities are split at a fixed ratio ($T_2 = 2T_1$ for undercoupled, and the mirror-swapped $T_1 = 2T_2$ for overcoupled), solved so the product R_1R_2 matches the critical cavity’s exactly. This is what guarantees the undercoupled and overcoupled fixtures at a given tier always share the same Finesse and the same on-resonance power dip, differing only in the phase behavior demonstrated below.

The reflected power dip alone cannot tell over- and undercoupled cavities apart: Table 3 shows that swapping R_1 and R_2 (which turns an undercoupled cavity into an overcoupled one, at the same Finesse, since Finesse depends only on the product r_1r_2) leaves $\min |r|^2$ completely unchanged. What differs is the *phase*. As detuning sweeps across one full FSR, $r(\varphi)$ traces an exact circle in the complex plane — a Möbius transform of the unit circle $e^{i\varphi}$,

so the image is provably a circle too, for any R_1, R_2 . Whether that circle encloses the origin, not merely how close it comes to it, is the real test.

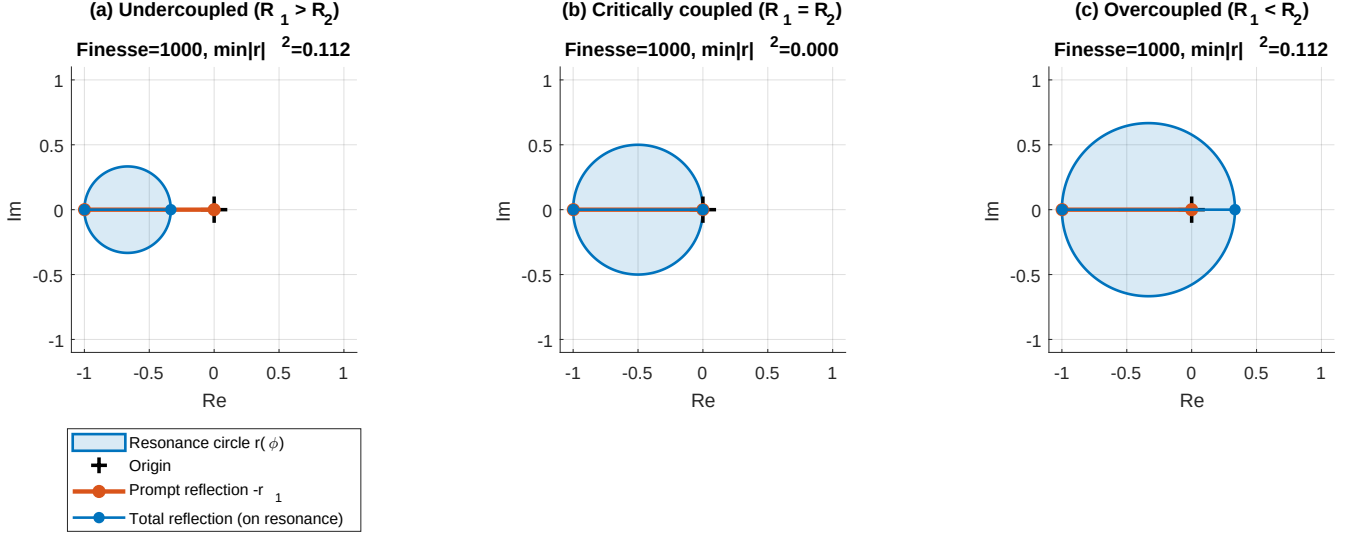


Figure 4: The reflection coefficient’s “resonance circle,” traced by $r(\varphi)$ over one full FSR, for an undercoupled, critically coupled, and overcoupled cavity (same product $r_1 r_2$, hence same Finesse, for panels (a) and (c) — only which mirror is called “front” has been swapped). The critically coupled circle (b) passes exactly through the origin. The undercoupled circle (a) stays entirely outside it. The overcoupled circle (c) encloses it. On-resonance reflection is decomposed with the same two colors as Figure 3: orange for the prompt reflection $-r_1$, blue for the total on-resonance reflection $r(0)$ it sums to (the net leaked-out contribution, without drawing each round trip) — note the blue endpoint sits on the *same* side in (a) but the *opposite* side in (c), despite both having identical $\min |r|^2$, and lands exactly on the origin in (b).

Regime	R_1	R_2	Finesse	$\min r ^2$	Net phase / FSR
Undercoupled	0.9979	0.9958	1000.0	0.1114	0°
Critically coupled	0.9969	0.9969	1000.0	0.0000	360°
Overcoupled	0.9958	0.9979	1000.0	0.1114	360°

Table 3: Minimum reflected power and net accumulated reflection phase over one full FSR sweep, for the three cavities of Figure 4. Generated by *generateReportFigures.m* (*exportCouplingRegimesTable*) directly from *OpticalCavity*.

Table 3’s last column makes the circle-enclosure argument concrete: the net phase accumulated by $r(\varphi)$ over one full FSR is exactly 360° whenever the resonance circle encloses or passes through the origin (critical and overcoupled), and exactly 0° — the phasor wobbles back and forth but never completes the loop — whenever the origin sits outside it (undercoupled). This is precisely the same 360° phase step already seen in Figure 2’s Panel 3, for the matched-mirror (critically coupled) running example.

To see this play out in the same panel layout as Figure 2 rather than only as an abstract circle, Figures 5 and 6 reuse *plotCavityResponse.m* directly for the *ExampleCavities.UndercoupledLowFinesse* and *ExampleCavities.OvercoupledLowFinesse* fixtures (§1.6 introduces the full *ExampleCavities* fixture set). The Low-Finesse tier

is used here so the phase's turning behavior is visible within the same ± 5 -linewidth window used throughout this report — the effect exists at any finesse, since Table 3 already demonstrated it numerically for the same fixtures' High-Finesse tier.

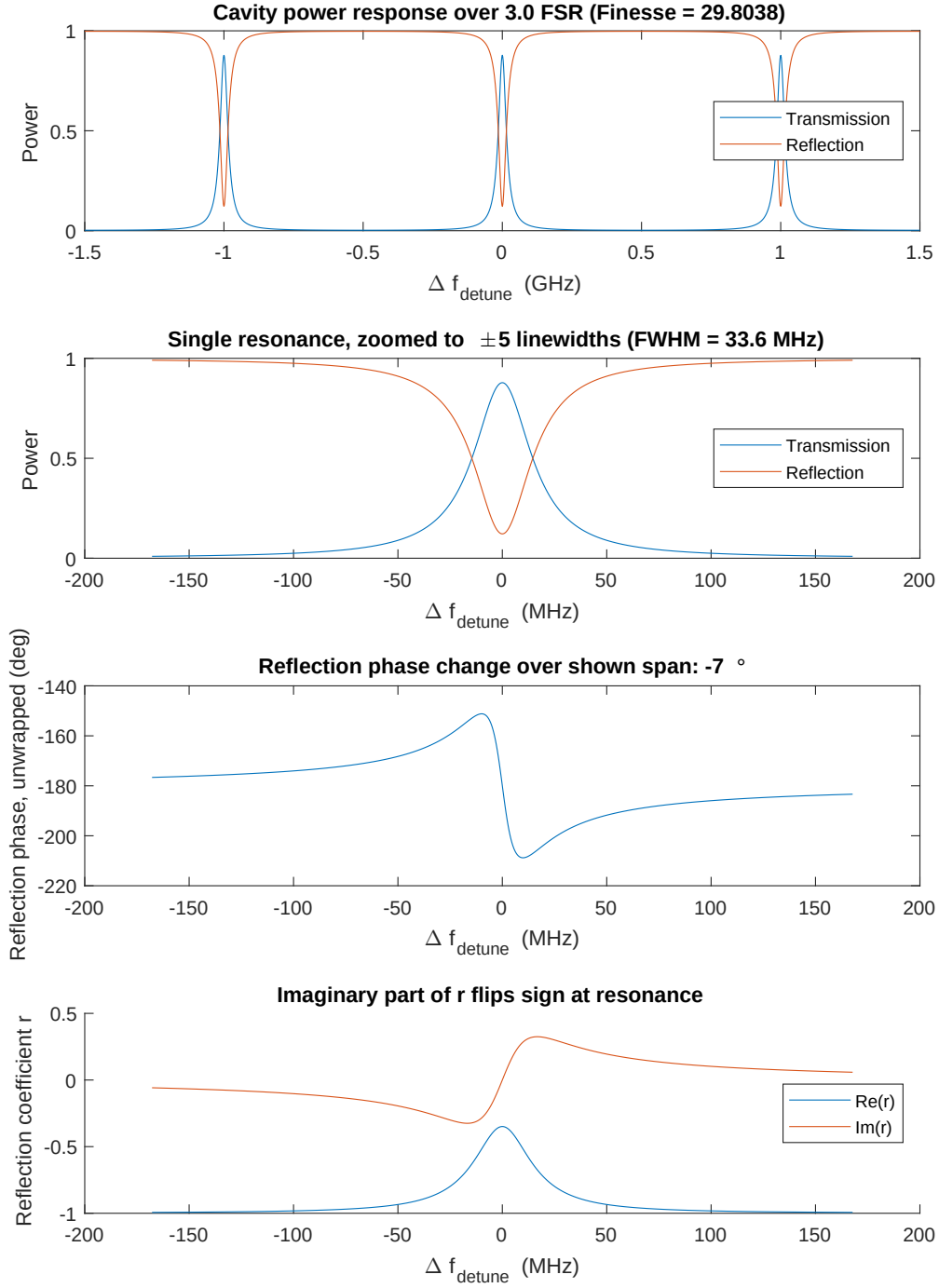


Figure 5: `plotCavityResponse.m` for *ExampleCavities.UndercoupledLowFinesse* ($R_1 = 0.9337 > R_2 = 0.8675$, $\mathcal{F} = 29.8$). Panel 3 shows the reflection phase swooping from -176.6° down to a minimum near -208.8° and turning back toward -183.4° — never approaching a monotonic 360° sweep; the net change over the shown span is only -6.7° , consistent with the 0° full-FSR winding number of Table 3. Panel 4's $\text{Re}(r)$ shows the corresponding small bump rather than a clean sign crossing, reflecting a reflection dip that comes close to zero ($\min |r|^2 = 0.1219$) but never reaches it.

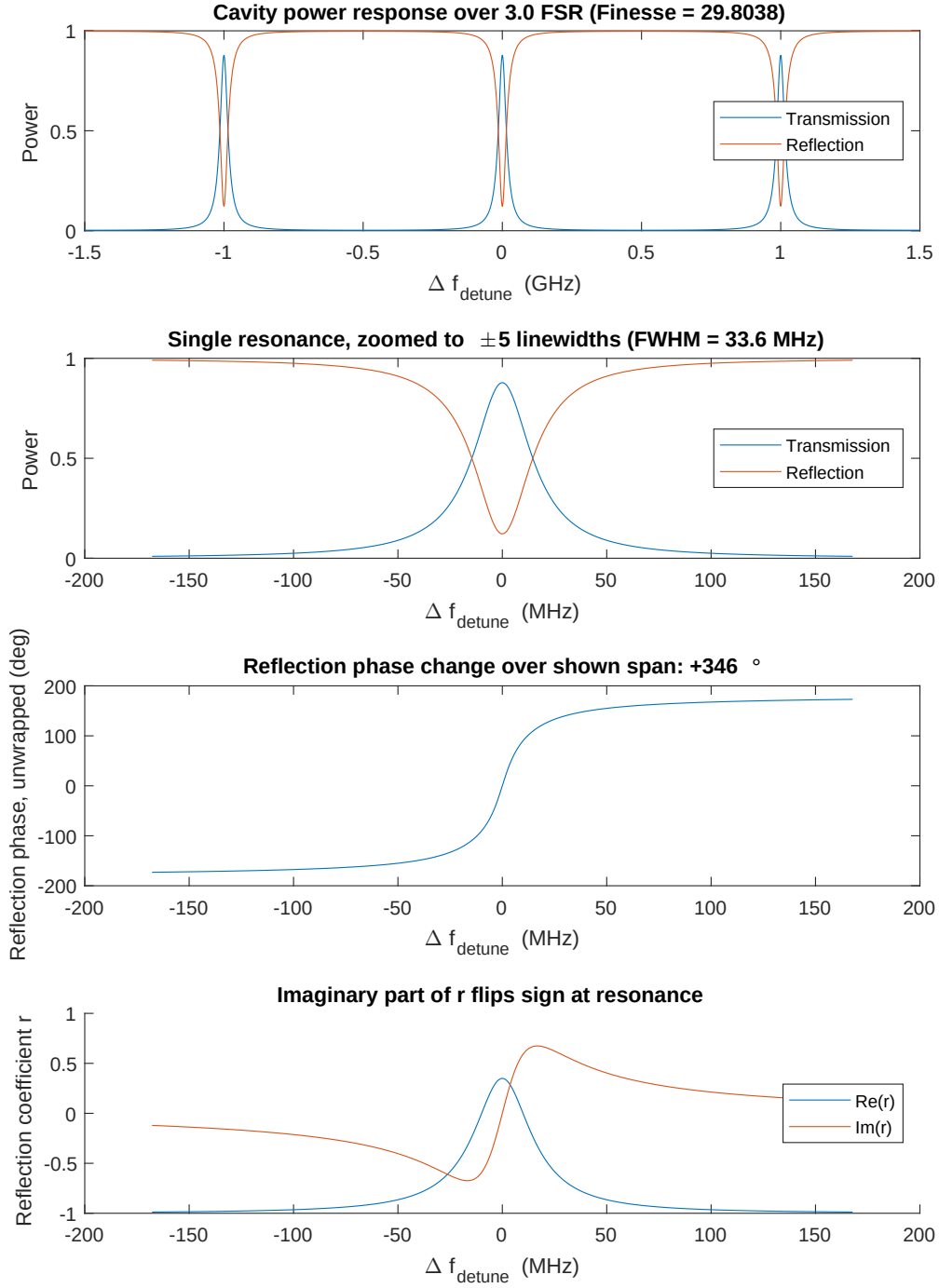


Figure 6: *plotCavityResponse.m* for *ExampleCavities.OvercoupledLowFinesse* — the mirror-swapped counterpart of Figure 5 above ($R_1 = 0.8675 < R_2 = 0.9337$, the same $\mathcal{F} = 29.8$ and the same $\min |r|^2 = 0.1219$). Panel 3 shows a clean, nearly monotonic sweep from -173.0° to $+173.0^\circ$, a net change of $+346.0^\circ$ — already most of a full 360° revolution within just ± 5 linewidths, confirming the resonance circle's enclosure of the origin. Panel 4's $\text{Re}(r)$ correspondingly crosses zero, unlike the undercoupled case above, even though the two cavities' reflection dips are identical in depth.

This distinction matters directly for PDH: the error signal built in Section 2 and later sections is derived entirely from this reflection phase response. An undercoupled cavity’s phase never completes its excursion, so the resulting error-signal feature is fundamentally weaker — a shallow wobble rather than a steep dispersive slope — even though its reflected-power dip can look identical to an overcoupled cavity’s. This is why cavities intended for PDH locking are designed to be critically coupled or overcoupled, never undercoupled: only those two regimes deliver the full, steep dispersive S-curve that Figure 2’s Panel 4 and everything built on it depend on.

2 Phase Modulation and Sidebands

2.1 Why modulate the light at all

The photodiode after the cavity can only measure power, the field’s squared magnitude. That throws away all phase information — and the whole point of PDH is to recover the cavity’s *phase* response (Section 1.4, Panels 3–4), which side of resonance the laser sits on, not just its amplitude response (symmetric, and unable to give a sign).

The trick: put a phase reference onto the light *before* it reaches the cavity, in the form of sidebands at a known frequency, chosen far outside the cavity linewidth. Those sidebands pass the cavity essentially unaffected (they are off-resonance), while the carrier — if it is close to resonance — picks up a cavity-induced phase shift. When carrier and sidebands recombine on the photodiode, they beat at that frequency, and that beat note’s phase encodes the carrier’s phase shift. Demodulating (mixing down) at the same frequency recovers it as a usable error signal. That demodulation step is the subject of a later ticket; this section is only about how the sidebands themselves come to exist.

2.2 The Jacobi-Anger expansion

Phase modulation imprints an oscillating phase term onto the carrier; its Jacobi-Anger expansion (below) turns that single oscillating phase into a discrete spectrum of sidebands. Table 4 collects the notation used throughout this section.

Symbol	Code identifier	Meaning
ω_{laser}	—	laser (carrier) angular frequency, $\omega_{\text{laser}} = 2\pi f_{\text{laser}}$ (the same f_{laser} of Eq. (3))
$E(t), E_0$	—	the modulated field, and its (unmodulated) carrier amplitude (Eq. (8))
β	<i>modulationIndex</i>	phase modulation depth, radians
ω_m, f_m	<i>modulationFrequencyHz</i>	modulation (angular) frequency
$J_\ell(\beta)$	<i>sidebands.Amplitude</i>	Bessel-function sideband amplitude at order ℓ
ℓ	<i>sidebands.Order</i>	sideband order, $\ell = -\text{MaxOrder} \dots \text{MaxOrder}$
$\mathcal{E}_\ell$	—	phasor amplitude of spectral line ℓ ($\ell = -1, 0, 1$), $\equiv J_\ell(\beta)$ (Eq. (12)); $\mathcal{E}_{-1} = -\mathcal{E}_1$
<i>IQ modulator (§2.5): subscript convention and arm labels</i>		
$(\cdot)_{I,Q}$	—	subscript convention, applied to several symbols below: “the same equation holds once with I throughout, once with Q throughout” — never a third, separate quantity of its own
I, Q (as arm labels)	—	positional labels for the two arms — never used bare for the demodulation-phase I/Q Quadrature choice of a later section (CONTEXT.md)
<i>IQ modulator: electric fields</i>		
$E_{\text{out}}(t)$	—	the IQ modulator’s own output field (Eq. (20))
$s_{I,Q}(t); s_I(t), s_Q(t)$	<i>inPhase, quadrature</i>	each arm’s own real, signed field-amplitude coefficient (dimensionless, normalized to E_0), defined by Eq. (16)/(18); <i>not</i> a voltage (contrast $V_{I,Q}(t)$ below)
<i>IQ modulator: biases and drive voltages</i>		
θ_{IQ}	—	the parent combiner’s own inter-arm bias (radians), nominally $\pi/2$, trimmed to compensate its manufacturing tolerance (Eq. (20))
$V_\pi, V_{\pi,I,Q}$	—	modulator half-wave voltage (Eq. (15)); the drive voltage that shifts one MZM’s phase by π ; generic, single-device form
$V_{\pi,I}, V_{\pi,Q}$	—	each arm’s own independently-set half-wave bias — Eq. (15) predicts them equal only if both arms’ crystal parameters match exactly, which real fabrication does not guarantee
$V_{I,Q}(t); V_I(t), V_Q(t)$	—	a single arm’s total drive voltage, bias plus RF (Eq. (17))
$v_{I,Q}(t); v_I(t), v_Q(t)$	—	the RF (small-signal) part of $V_{I,Q}(t)$ layered on top of its bias: $v_{I,Q}(t) = V_{I,Q}(t) - V_{\pi,I,Q}$
<i>IQ modulator: Pockels cell (material/geometry constants)</i>		
$n_e, r_{33}, d, L_{\text{mod}}$	—	extraordinary refractive index, electro-optic coefficient, electrode gap, interaction length (Eq. (13)) — L_{mod} deliberately not written L , to avoid colliding with Section 1’s cavity length
$\Delta n_e(V)$	—	Pockels-effect perturbation of n_e under a generic applied voltage V (Eq. (13))
$\Delta\phi(V)$	—	voltage-controlled phase shift of a single MZM sub-path under V (Eq. (14))

Table 4: Notation used throughout Section 2.

The field leaving the electro-optic modulator (EOM) is

$$E(t) = E_0 e^{i(\omega_{\text{laser}} t + \beta \sin \omega_m t)}, \quad (8)$$

where $\omega_{\text{laser}} = 2\pi f_{\text{laser}}$ is the laser's own angular frequency — the same f_{laser} introduced alongside detuning in Eq. (3), just in angular rather than linear units; it is deliberately not written ω_0 , to avoid colliding with Section 1's unrelated use of the subscript 0 for a resonance's mode number ($f_{0,N}$, Eq. (2)). With modulation index β (how many radians of phase are being wobbled) and modulation angular frequency $\omega_m = 2\pi f_m$, factoring out the fast carrier oscillation and expanding what is left using the Jacobi-Anger identity,

$$e^{i\beta \sin \omega_m t} = \sum_{\ell=-\infty}^{\infty} J_{\ell}(\beta) e^{i\ell \omega_m t}, \quad (9)$$

which is an *identity*, not an approximation — the approximation only enters when the sum is truncated. Each term ℓ is a sideband at frequency $\omega_{\text{laser}} + \ell \omega_m$, weighted by the Bessel function $J_{\ell}(\beta)$. *generatePhaseModulationSidebands.m* computes exactly this table of coefficients (`besselj(order, modulationIndex)`) for $\ell = -\text{MaxOrder} \dots \text{MaxOrder}$.

Keeping only $\ell \in \{-1, 0, +1\}$ (the project's *MaxOrder* = 1 default):

$$E(t) \approx E_0 [J_0(\beta) + J_1(\beta) e^{i\omega_m t} + J_{-1}(\beta) e^{-i\omega_m t}] e^{i\omega_{\text{laser}} t}. \quad (10)$$

The identity $J_{-1}(\beta) = -J_1(\beta)$ (odd-order Bessel functions are odd functions of their order) is not a detail — it is the entire reason phase modulation looks different from amplitude modulation in the phasor picture. Substituting it into Eq. (10) and using Euler's formula:

$$J_0(\beta) + J_1(\beta) [e^{i\omega_m t} - e^{-i\omega_m t}] = J_0(\beta) + i 2J_1(\beta) \sin(\omega_m t). \quad (11)$$

The real part stays pinned at $J_0(\beta)$; the imaginary part swings back and forth as $2J_1(\beta) \sin(\omega_m t)$. In the complex plane, the resultant phasor does not trace a circle (that would be amplitude modulation) — it wiggles up and down along a vertical line centered on the real axis. That vertical wiggle *is* phase modulation: since $\arg(E) \approx \frac{2J_1(\beta)}{J_0(\beta)} \sin(\omega_m t)$ for small deviations, the phase angle oscillates while the magnitude stays almost exactly $J_0(\beta)$ — exactly constant only if the full infinite sum of Eq. (9) is kept. At $\beta = 1.08$, $J_0(1.08) = 0.7281$, $J_1(1.08) = 0.4656$, and truncating at $\ell = \pm 1$ reproduces the exact field of Eq. (8) to within about 27% at the extremes of the cycle — the size of the next dropped term, $J_2(1.08) \approx 0.1326$, doubled for the pair $\ell = \pm 2$, considerably rougher than the truncation would be at a smaller modulation depth. Section 2.3 returns to why this particular β is used here, and to the higher-order sidebands this truncation drops.

2.3 Amplitude scaling with modulation depth

Eq. (9) says more than “there are sidebands”: each sideband's amplitude $J_{\ell}(\beta)$ is a specific, computable function of the modulation index, so raising or lowering β redistributes field amplitude between the carrier and the sidebands in a definite way, not just a vague “more modulation, more sidebands.” Figure 7 plots the carrier and the $\ell = \pm 1, \pm 2, \pm 3$ sidebands as vertical lines at four representative values of β (*ExampleModulationRegimes*), spanning what this project calls the Modulation Regimes.

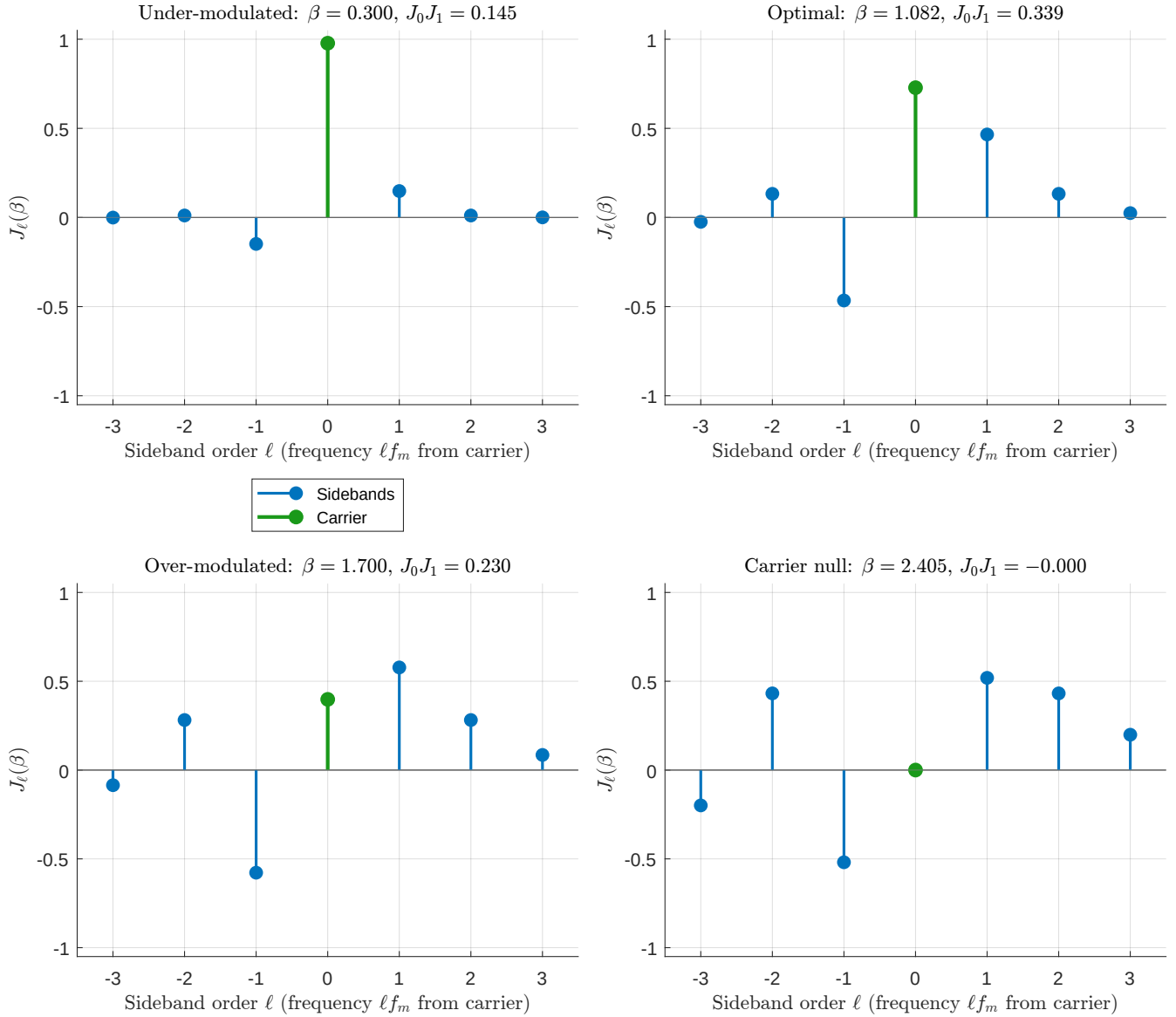


Figure 7: Carrier (green) and sideband (blue) field amplitudes $J_\ell(\beta)$ vs. sideband order ℓ , at the four *ExampleModulationRegimes* values. Under-modulated ($\beta = 0.3$): almost all the field stays in the carrier. Optimal ($\beta \approx 1.08$, this project’s running example): the carrier/sideband amplitude product $J_0(\beta)J_1(\beta)$ is maximized. Over-modulated ($\beta = 1.7$): field has moved past the optimum into higher-order sidebands, and the product has fallen again. Carrier null ($\beta = 2.4048$, the first zero of J_0): the carrier line vanishes entirely.

At $\beta = 0.3$ (**under-modulated**), the carrier line towers over sidebands that are barely visible above the axis: most of the field amplitude never left the carrier. Raising β moves amplitude out of the carrier and into the sidebands — visibly so between the first two panels — but not without limit: past $\beta \approx 1.08$ (**over-modulated**), the carrier keeps shrinking while the amount landing in the $\ell = \pm 1$ sidebands specifically (as opposed to higher orders) starts shrinking too, spreading out into $\ell = \pm 2, \pm 3$ instead. Pushed far enough, at $\beta = 2.4048$ the carrier line disappears

altogether (**carrier null**, the first zero of J_0) — all of the original carrier amplitude has been redistributed among the sidebands.

The value $\beta \approx 1.08$ rad already used, unexplained, in the worked example of §2.2 is not arbitrary. Why does it matter specifically, rather than “more sidebands are always better”? Section 2.1 already named the mechanism the rest of this report builds on: carrier and sidebands eventually recombine on a photodiode and *beat* against each other, and it is that beat note’s phase that carries the cavity’s response. A beat note needs both parties present in some amount — an undetectably small sideband has nothing to beat against, however cleanly phase-shifted the carrier becomes, and an undetectably small carrier is equally useless however strong the sidebands are. That is why the relevant quantity is not $J_1(\beta)$ alone (which keeps growing well past $\beta = 1.08$, not peaking until $\beta \approx 1.8$) but the *product* $J_0(\beta)J_1(\beta)$: it has an interior maximum precisely because growing one factor eventually costs more in the other than it gains. This report does not yet derive the demodulated PDH error signal itself (that is the subject of the next sections), but the same product sets its slope, which is why $\beta \approx 1.08$ rad — maximizing $J_0(\beta)J_1(\beta)$ at ≈ 0.339 , more than double the under-modulated regime’s 0.145 — is the standard choice quoted in the literature [1] for getting the strongest possible PDH signal out of a given amount of laser power.

That maximum, though, is a maximum of the $\ell = \pm 1$ content alone — and §2.2’s worked example already showed that at $\beta \approx 1.08$ the $\ell = \pm 1$ truncation itself is a rough approximation, because $\ell = \pm 2$ and higher are no longer negligible (Figure 7, third and fourth panels). Is there a way to modulate the carrier without also populating these higher orders? Not with a single EOM driven at one tone: Eq. (9) is an identity, so for any nonzero β every order ℓ has a nonzero $J_\ell(\beta)$, and there is no separate knob that turns off $\ell = \pm 2$ while leaving $\ell = \pm 1$ alone — β is the only free parameter, and it sets the whole spectrum at once. What does change with β is how much of the total field ends up beyond first order: for small β , $J_\ell(\beta) \approx \frac{1}{\ell!}(\beta/2)^{|\ell|}$, so each higher order is suppressed by an extra power of β relative to the last. The ratio $J_2(\beta)/J_1(\beta)$ makes this concrete: 7.5% at the under-modulated $\beta = 0.3$, 28.5% at the optimal $\beta \approx 1.08$, and 48.8% at the over-modulated $\beta = 1.7$ — which is exactly why real PDH setups often deliberately run somewhat below the strict $J_0(\beta)J_1(\beta)$ optimum, trading a little error-signal slope for a spectrum where the ignored higher orders stay small.

The higher orders are not, on their own, harmful to the eventual error signal the way an over-modulated carrier is. Since f_m is chosen far outside the cavity linewidth (§2.1), $\ell = \pm 2, \pm 3, \dots$ sit even further from resonance than $\ell = \pm 1$ and come back from the cavity essentially as untouched as the first-order sidebands do. What they cost is optical power diverted away from the carrier and first-order sidebands that actually carry the usable beat note, plus extra beat terms between adjacent higher orders that add a roughly detuning-independent background to the demodulated signal rather than a resonance feature — a nuisance to filter around, not a corruption of the signal’s shape (the full argument needs the demodulation math of the next sections). A dedicated narrowband optical filter placed after the EOM could reject them outright, since they sit at clean integer multiples of f_m , but this is rarely done in practice — backing off β a little is simpler, and, as the ratios above show, already suppresses them geometrically.

2.4 The phasor picture

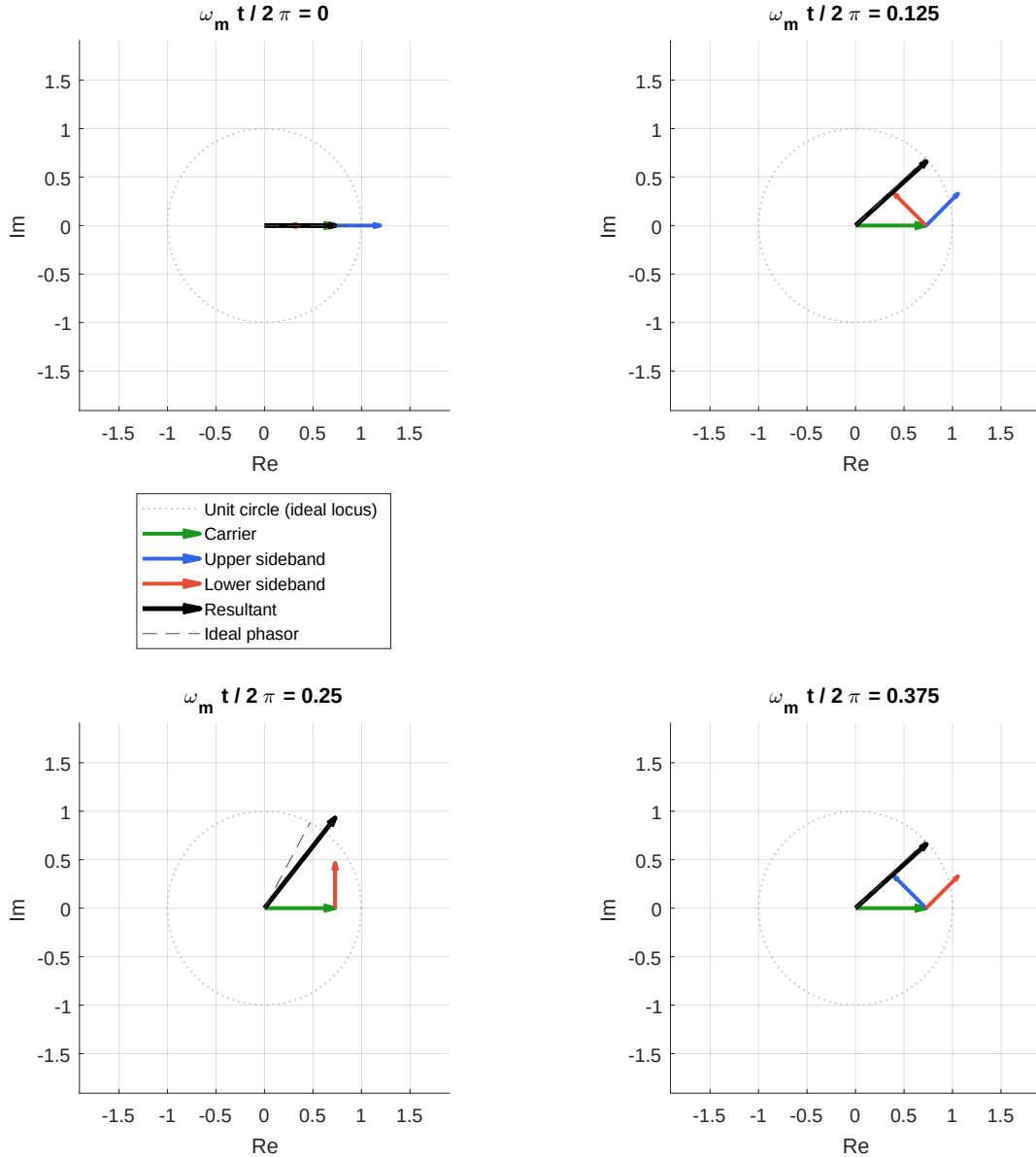


Figure 8: Four snapshots across one modulation cycle of the carrier + sideband phasor picture (a static stand-in for the live animation in *animatePhaseModulatedPhasors.m*), at $\beta \approx 1.08$ (Optimal Modulation Depth, §2.3), $f_m = 15$ MHz. Green: carrier, fixed (the frame co-rotates with it). Blue/red: upper/lower sideband, drawn starting at the carrier’s tip, rotating at $\pm\omega_m$. Black: the true resultant. Dashed grey: the exact (infinite-order) phasor $e^{i\beta \sin \omega_m t}$ at this instant. Dotted light grey: the full unit circle that exact phasor traces over the whole cycle — radius exactly 1, since pure phase modulation changes only $\arg(E)$, never $|E|$.

Figure 8 draws exactly the three-phasor picture of Eq. (10), in the frame rotating with the carrier (so the carrier itself sits still). The sideband arrows are drawn starting at the tip of the carrier arrow, not at the origin, so their

vector sum relative to that tip is visible directly as the thing that makes the resultant wobble.

Name each arrow's own complex amplitude explicitly, since later sections (the cavity's effect on each line, §3.2, and the dot-product construction of §4) need to refer to it directly rather than re-deriving it from the Jacobi-Anger expansion each time:

$$\mathcal{E}_\ell \equiv J_\ell(\beta), \quad \ell \in \{-1, 0, 1\}, \quad (12)$$

the carrier's own phasor amplitude $\mathcal{E}_0$ (green) and the upper/lower sideband phasor amplitudes $\mathcal{E}_{\pm 1}$ (blue/red) of Eq. (10) — with $\mathcal{E}_{-1} = -\mathcal{E}_1$ inherited directly from $J_{-1}(\beta) = -J_1(\beta)$, the same odd-order identity already used to reach Eq. (11).

The light grey dotted circle is not a visual approximation: pure phase modulation changes only the phasor's angle, never its length, so the exact field $E(t)/E_0 = e^{i\beta \sin \omega_m t}$ has $|E(t)/E_0| = 1$ for every t , exactly, regardless of β — because $|e^{i\theta}| = 1$ for any real θ , and $\beta \sin \omega_m t$ is always real. Amplitude modulation would instead change the radius and leave the angle alone; phase modulation does the opposite, which is why the exact phasor traces this particular circle, of radius exactly 1 and not merely close to it, rather than some other curve.

Two things to check across the four panels:

1. **The counter-rotation itself.** At every instant, the upper sideband's angle and the lower sideband's angle are reflections of each other through the horizontal: one is $e^{+i\omega_m t}$, the other $e^{-i\omega_m t}$ — same speed, opposite direction. This is a direct consequence of $J_{-1}(\beta) = -J_1(\beta)$; a symmetric spectrum ($J_{-1} = +J_1$, as in amplitude modulation) would instead have both sidebands co-rotating.
2. **Why counter-rotation turns into a vertical wobble, not a circular motion.** At $t = 0$ (top-left panel) both sideband arrows point along the real axis in opposite directions and exactly cancel: the resultant sits right on the carrier's tip, no wobble yet. As t advances (remaining panels), the two arrows swing away from the real axis in mirrored directions, so their real components stay matched (still canceling) while their imaginary components both grow the same way — precisely Eq. (11).

At this modulation depth the resultant (black) departs visibly from the dotted unit circle in both directions across the wobble — dipping inside to $|J_0(\beta)| \approx 0.73$ at $t = 0$ (top-left panel) and bulging outside to ≈ 1.18 at the wobble's peak (bottom-left panel) — instead of merely grazing it the way a smaller β would. Both excursions are the same $\approx 27\%$ truncation error from keeping only $\ell = 0, \pm 1$, the one quantified numerically at the end of Section 2.2.

2.5 Synthesizing the sidebands with an IQ modulator

Section 2.3 asked whether the higher orders could be switched off without also killing $\ell = \pm 1$, and found no such knob on a single EOM: β is its only free parameter, and Eq. (9) being an identity means it sets every order at once. An *IQ modulator* answers the same question differently, by replacing one nonlinearly-driven waveguide with two independently-biased ones, combined in quadrature.

This subsection derives $E_{\text{out}}(t)$ as an explicit function of five independent inputs — three biases ($V_{\pi,I}$, $V_{\pi,Q}$, θ_{IQ}) and two time-varying drive voltages ($v_I(t)$, $v_Q(t)$) — plus the Pockels cell's own material/geometry constants (n_e , r_{33} , d , L_{mod} , and λ shared with Section 1). Every intermediate quantity along the way gets its own equation, in order, so the chain from those inputs to $E_{\text{out}}(t)$ has no unstated step.

The physics: a crystal, not a circuit. Both a phase-modulating EOM (Eq. (8)) and an IQ modulator exploit the same underlying effect: the linear electro-optic (Pockels) effect in a non-centrosymmetric crystal such as lithium niobate (LiNbO_3). Applying a voltage V across electrodes patterned alongside a waveguide linearly perturbs the crystal's own (extraordinary) refractive index,

$$\Delta n_e(V) = -\frac{1}{2}n_e^3 r_{33} \frac{V}{d}, \quad (13)$$

where r_{33} is the relevant electro-optic coefficient and d the electrode gap — solid-state physics of a specific kind: a bulk dielectric crystal's own birefringence, tuned by an applied field, with no free carriers or semiconductor band

structure involved (unlike, say, an electro-absorption modulator), which is also why it is fast: nothing has to move except the light itself. Over an interaction length L_{mod} , Eq. (13) becomes a voltage-controlled phase shift,

$$\Delta\phi(V) = \frac{2\pi}{\lambda} \Delta n_e(V) L_{\text{mod}}, \quad (14)$$

and the half-wave voltage V_π — the drive voltage that shifts phase by exactly π — is defined by $|\Delta\phi(V_\pi)| = \pi$ in Eq. (14), i.e.

$$V_\pi = \frac{\lambda d}{n_e^3 r_{33} L_{\text{mod}}}. \quad (15)$$

Eq. (14) is exactly what a phase-modulating EOM applies directly to the beam, with $V(t) \propto \sin \omega_m t$ reproducing Eq. (8)’s $\beta \sin \omega_m t$. An IQ modulator instead uses it as a building block, one level down — and, since a device’s two arms are two physically separate crystal regions, Eq. (15) predicts the *same* V_π for both only if n_e , d , L_{mod} match between them exactly. Real fabrication does not hold that tightly, which is exactly why $V_{\pi,I}$ and $V_{\pi,Q}$ (below) are supplied as two independently-set bias inputs rather than computed from Eq. (15) at run time.

From phase shift to intensity: one child Mach-Zehnder modulator per arm. A single Mach-Zehnder modulator (MZM) splits the light into two sub-paths, drives them push-pull (one gets $+V(t)/2$, the other $-V(t)/2$, so no common phase — no chirp — survives the recombination), and interferes them back together. Writing this once for either arm (subscript I, Q meaning “the same equation holds with I throughout, and again with Q throughout”), the relative phase from Eq. (14) becomes a real, signed field-amplitude coefficient

$$s_{I,Q}(t) \equiv \cos\left(\frac{\pi V_{I,Q}(t)}{2V_{\pi,I,Q}}\right), \quad (16)$$

a nonlinear “raised-cosine” transfer curve in the arm’s own total drive voltage $V_{I,Q}(t)$ — bias plus RF, defined next.

Biasing at the null: where “bias” and “RF in” come from. Each arm’s total drive voltage is its own fixed bias plus a time-varying RF part,

$$V_{I,Q}(t) = V_{\pi,I,Q} + v_{I,Q}(t), \quad (17)$$

i.e., explicitly, $V_I(t) = V_{\pi,I} + v_I(t)$ and $V_Q(t) = V_{\pi,Q} + v_Q(t)$. Substituting Eq. (17) into Eq. (16) — biasing at the transfer curve’s own field null, $V_{I,Q} = V_{\pi,I,Q}$, not the power-quadrature point $V_{\pi,I,Q}/2$ sometimes used for unipolar analog links — gives an *exact* identity, not an approximation:

$$s_{I,Q}(t) = \sin\left(\frac{\pi v_{I,Q}(t)}{2V_{\pi,I,Q}}\right) \approx \frac{\pi v_{I,Q}(t)}{2V_{\pi,I,Q}} \quad (|v_{I,Q}(t)| \ll V_{\pi,I,Q}), \quad (18)$$

linear (in the small-signal limit only), bipolar, and passing through zero at $v_{I,Q} = 0$ — exactly the shape needed to represent a signed real drive amplitude; the sign shown is a choice of which electrode is driven positive, flipped by wiring the electrodes the other way. $V_{\pi,I,Q}$ is the *bias* Figure 9 labels; $v_{I,Q}(t)$ is the *RF in* next to it, superimposed on top. Written out explicitly for each arm on its own — the two equations this subsection is named for —

$$s_I(t) = \sin\left(\frac{\pi v_I(t)}{2V_{\pi,I}}\right), \quad s_Q(t) = \sin\left(\frac{\pi v_Q(t)}{2V_{\pi,Q}}\right). \quad (19)$$

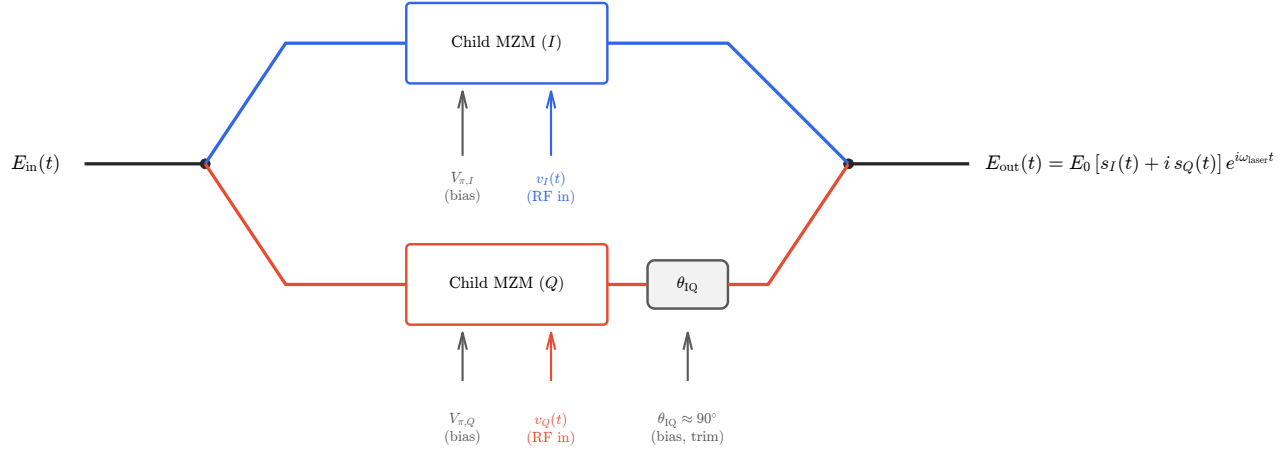


Figure 9: Dual-parallel Mach-Zehnder IQ modulator, schematically. Input light splits into an I -arm (blue) and a Q -arm (red), each carrying its own child MZM (Eq. (16)) with its own bias $V_{\pi,I,Q}$ and RF input $v_{I,Q}(t)$ (Eqs. (17)–(19)); the Q -arm then passes through a third, independently trimmable bias $\theta_{IQ} \approx 90^\circ$ (compensating the parent combiner’s own manufacturing tolerance, not shown to scale) before the two arms recombine (Eq. (20)).

Two arms in quadrature: the complex output field. The two arms’ field-amplitude coefficients $s_I(t)$, $s_Q(t)$ (Eq. (19)) recombine through a *third*, independent bias, θ_{IQ} , applied to the Q -arm alone:

$$E_{\text{out}}(t) = E_0 [s_I(t) + e^{i\theta_{IQ}} s_Q(t)] e^{i\omega_{\text{laser}} t}, \quad \theta_{IQ} = \frac{\pi}{2} \text{ (nominal)}, \quad (20)$$

absorbing the arms’ static splitting/recombining loss into E_0 ’s own normalization, the same way this report already leaves such constant prefactors out of E_0 elsewhere. Unlike $V_{\pi,I}/V_{\pi,Q}$ (which each linearize *one arm’s own* amplitude response, Eq. (18)), θ_{IQ} sets the angle *between* the two arms and carries no amplitude information at all. It is also not fixed by geometry the way the nominal value $\pi/2$ suggests: a parent splitter/combiner is never fabricated to exactly 3 dB and exactly 90° , so a real device trims θ_{IQ} electrically until the two arms measure in quadrature, compensating that residual manufacturing tolerance — three independent bias controls in total ($V_{\pi,I}$, $V_{\pi,Q}$, θ_{IQ}), each doing a different job, even though casual usage calls all three simply “the bias.” At the nominal $\theta_{IQ} = \pi/2$ ($e^{i\pi/2} = i$), Eq. (20) reduces to $E_0[s_I(t) + i s_Q(t)]e^{i\omega_{\text{laser}} t}$, exactly the general in-phase/quadrature decomposition of a modulated carrier: any real $s_I(t)$, $s_Q(t)$ pair is reachable, independently of each other, which is the whole point of building two child modulators instead of driving one.

It is worth being precise about what actually oscillates where. $v_Q(t)$ modulates a *phase* inside the Q -arm’s own two sub-paths (Eq. (14)) — but by the time that arm’s light recombines at its *own* child MZM’s output, what emerges is a real, signed *amplitude* $s_Q(t)$ (Eq. (19)), not an oscillating optical phase. The Q -arm’s phase *relative to the I-arm* is a separate thing, set once by θ_{IQ} above: it does not itself track $\sin(\omega_m t)$, and is not the quantity this subsection modulates.

Choosing the drive: reproducing Eq. (11) exactly. Set $v_I(t)$ to a pure DC bias (no RF at all) and $v_Q(t)$ to a single sinusoidal tone at f_m , tuned via Eq. (19) so that

$$s_I(t) = J_0(\beta), \quad s_Q(t) = 2J_1(\beta) \sin(\omega_m t). \quad (21)$$

Substituting Eq. (21) into Eq. (20) at the nominal $\theta_{IQ} = \pi/2$ gives $E_0[J_0(\beta) + i 2J_1(\beta) \sin(\omega_m t)]e^{i\omega_{\text{laser}} t}$ — Eq. (11) exactly, by inspection. It is worth expanding that $\sin(\omega_m t)$ back out, though, since “only a single RF tone was applied” can look like it should give only one sideband, not two. Euler’s formula splits any *real* oscillation into two complex exponentials, never one:

$$i 2J_1(\beta) \sin(\omega_m t) = i 2J_1(\beta) \cdot \frac{e^{i\omega_m t} - e^{-i\omega_m t}}{2i} = J_1(\beta) e^{i\omega_m t} + J_{-1}(\beta) e^{-i\omega_m t}, \quad (22)$$

using $J_{-1}(\beta) = -J_1(\beta)$ in the last step — so Eq. (21) substituted into Eq. (20) is, term by term, exactly Eq. (10)’s $J_0(\beta) + J_1(\beta)e^{i\omega_m t} + J_{-1}(\beta)e^{-i\omega_m t}$: both the upper *and* the lower sideband, even though $v_Q(t)$ only ever carried one RF tone at $+f_m$. The $-\omega_m$ term is not something the IQ modulator has to synthesize separately — it is the unavoidable mirror half of $\sin(\omega_m t)$ ’s own complex-exponential decomposition, present because $s_Q(t)$ is a *real* number (Eq. (19)), the same reason §2.4’s phasor picture needed two counter-rotating arrows, not one, to draw a single real phase wobble. A device that could apply a genuinely *complex* drive voltage could suppress one sideband (single-sideband modulation) — but no physical voltage source produces one; that would need a second, RF-domain IQ stage combining two RF tones 90° apart, a different trick one level up from the optical IQ modulator derived here. Eq. (21)–(22) reproduce Eq. (11)/(10) exactly — but now as the literal content of what was fed in, not as a truncation of a richer spectrum the way Eq. (9) needs truncating: a DC value and a single sinusoid have no other frequency to give. That claim is about the idealized envelope abstraction of Eq. (20) alone, though — the same level of idealization this report already uses elsewhere (a lossless cavity, Section 1); the physical crystal only reaches it to the extent each child modulator’s $\sin(\cdot)$ stays in Eq. (18)’s small-signal regime. Push $v_Q(t)$ ’s amplitude far enough and that same $\sin(\cdot)$ nonlinearity reintroduces a Jacobi-Anger-type harmonic series one level down — Eq. (9)’s mathematics again, now happening inside the modulator rather than on the beam it drives. *generateIqModulatorDrive.m* models the idealized abstraction, cross-checked in *generateIqModulatorDriveTest.m* against Section 2.2’s truncated sum with zero energy at $|\ell| \geq 2$; the crystal-level linearization above is discussed for physical grounding, not separately modeled numerically.

At this project’s Optimal Modulation Depth ($\beta \approx 1.08$, §2.3), $s_I(t) = J_0(\beta) = 0.7281$ (a pure DC bias) and $s_Q(t) = 2J_1(\beta) \sin(\omega_m t)$ swings between ± 0.9312 (Figure 10, top panel) — the same numbers already computed in §2.2. The resulting spectrum (bottom panel, an exact one-cycle FFT of the reconstructed envelope, not assumed) has all of its energy at $\ell = 0, \pm 1$ and none anywhere else, to within floating-point round-off ($\sim 10^{-16}$) — contrast Figure 7’s own optimal- β panel, which shows real, nonzero energy at $\ell = \pm 2, \pm 3$ from the same β on a single-tone-driven EOM.

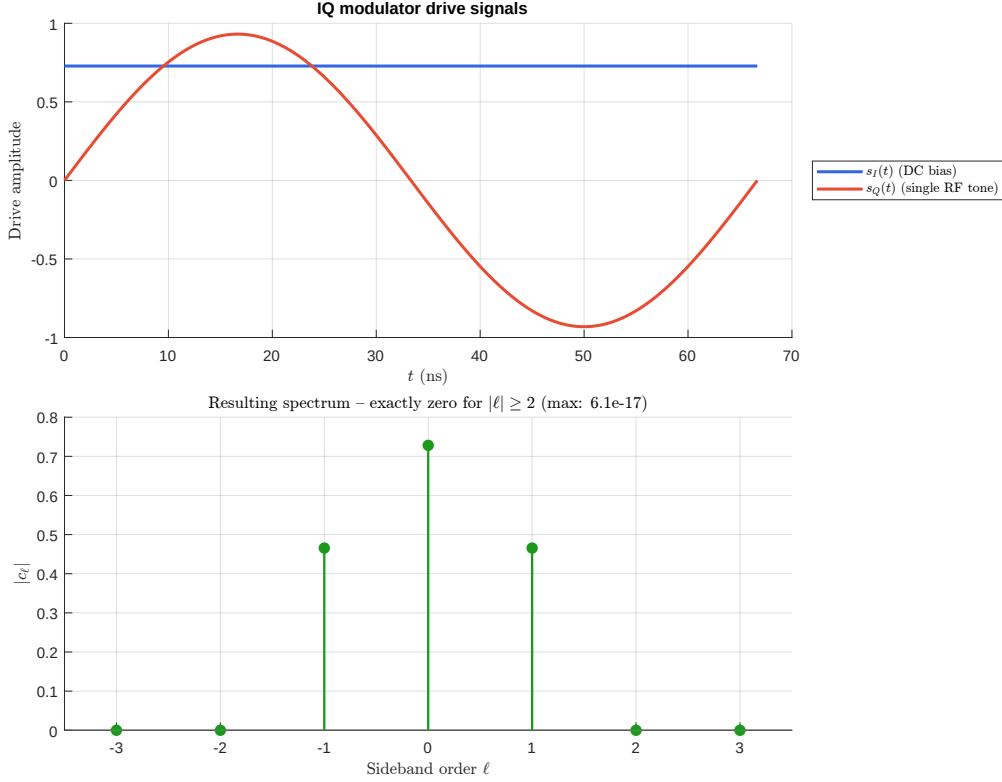


Figure 10: Top: the IQ modulator drive signals $s_I(t)$, $s_Q(t)$ (*generateIqModulatorDrive.m*) at $\beta = \text{Optimal Modulation Depth}$, $f_m = 15 \text{ MHz}$ — a DC bias and a single RF tone. Bottom: the resulting spectrum, recovered by an FFT over one modulation cycle of the reconstructed envelope $s_I(t) + i s_Q(t)$; exactly three lines, no higher orders, unlike Figure 7’s EOM spectrum at the same β .

2.6 Where this leads

Once the beam described by Eq. (10) enters the cavity of Section 1, each of its three frequency components sees its own detuning from the nearest cavity resonance (Eq. (3)) — Δf_{detune} for the carrier, $\Delta f_{\text{detune}} + f_m$ and $\Delta f_{\text{detune}} - f_m$ for the upper and lower sideband — and so picks up its own value of the cavity’s reflection coefficient (Eq. (7)). If f_m is far outside the cavity linewidth, the two sidebands return essentially untouched while the carrier alone is distorted near resonance. Extracting that distortion — the photodiode/mixer/filter detection chain, and the phasor dot-product shortcut that reproduces it without simulating time at all — is the subject of the next sections of this report, to be added as the corresponding tickets of the underlying wayfinder map are worked through.

3 The PDH Detection Chain

3.1 The full locking setup

Section 2.6 left off with a beam — carrier plus two sidebands, either from an EOM (§2.2) or an IQ modulator (§2.5) — reflecting off the cavity of Section 1, each of its three lines picking up its own value of the cavity’s reflection coefficient. This section extracts a usable error signal from that reflected light: the photodiode, mixer, and low-pass filter that turn an optical field back into a single number a servo could act on.

servo + PZT (not modeled --- see phase 2)

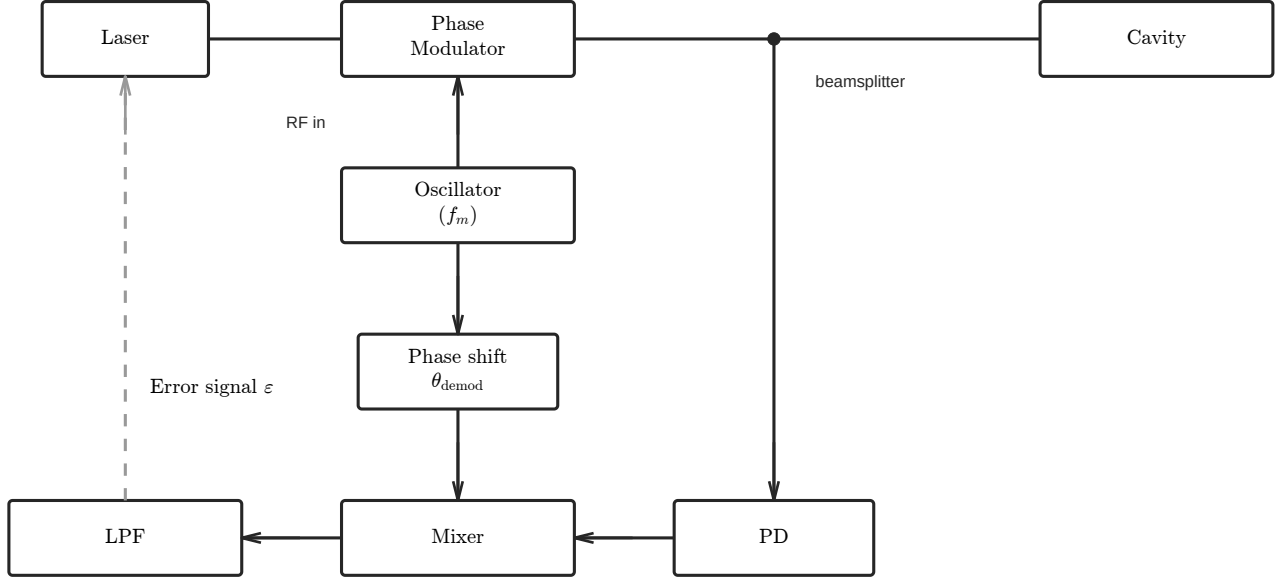


Figure 11: The full PDH laser-locking setup, schematically. The phase modulator is drawn generically — either §2.2’s EOM or §2.5’s IQ modulator can fill this role, since nothing downstream depends on which one produced the sidebands. A single oscillator at f_m — the local oscillator, or LO, in the mixer’s usual sense — drives the modulator directly and, through an adjustable phase shifter (the demodulation phase, named in Section 3.2), the mixer’s other input. The servo and PZT that would close the loop back onto the laser (grey, dashed) are not modeled by this report — see the “Kick off phase-2 map” ticket of the underlying wayfinder map — this section stops at the error signal itself.

Two things are worth naming before the math. First, the beamsplitter that picks off the reflected beam for the photodiode is drawn as a plain, generic element: this report has never modeled polarization, and a real setup’s polarizing-beamsplitter-plus-quarter-wave-plate isolation (keeping reflected light out of the laser) would add a category of detail this project doesn’t otherwise track. Second, the single oscillator driving both the modulator and (through its own phase shifter) the mixer is what makes the demodulation phase meaningful at all: it is a phase *relative to* the modulation itself, not an absolute one, which is exactly why Section 3.3 can tune it independently of everything else in the chain.

3.2 From reflected field to photodiode power

Table 5 collects the notation new to this section — everything else ($J_\ell(\beta)$, $r(\varphi)$, f_m , Δf_{detune} , FSR) reuses Sections 1 and 2 exactly as already defined.

Symbol	Code identifier	Meaning
k	—	harmonic-order label, $k \equiv \ell - \ell' \in \{-2, -1, 0, 1, 2\}$, introduced by grouping Eq. (26)'s terms — distinct from Section 1's round-trip index k of Eqs. (6)–(7) [forward-reference: that round-trip k is a different, unrelated symbol that happens to share the letter, used earlier in the document; this table's own harmonic-order k is not used before this point]
φ_ℓ	—	per-line round-trip phase, extending Section 1's φ to each sideband's own detuning (Eq. (23))
$\mathcal{E}_{\text{refl},\ell}$	—	line ℓ 's own reflected phasor, $\equiv \mathcal{E}_\ell r(\varphi_\ell)$ (Eq. (24)) — extends §2.4's phasor amplitude $\mathcal{E}_\ell$ through the cavity
$E_{\text{refl}}(t)$	—	the cavity-reflected field's baseband envelope, carrier-frame (Eq. (25))
$P_{\text{PD}}(t)$	—	the photodiode's total power (Eq. (26))
$P_{\text{PD},\text{DC}}$	<i>dcReflectedPower</i>	the DC part of $P_{\text{PD}}(t)$ (Eq. (28))
$P_{\text{PD},\omega_m}(t), P_{\text{PD},2\omega_m}(t)$	<i>computeReflectedPowerFrequencyScan</i>	its $\omega_m/2\omega_m$ parts (Eqs. (30), (32))
C	—	the beat-note coefficient at ω_m (Eq. (29))
D	—	the beat-note coefficient at $2\omega_m$ (Eq. (31))
θ_{demod}	<i>DemodulationPhase</i>	the LO's own reference phase (Eq. (33)) — deliberately not bare θ , to stay visually distinct from §2.5's unrelated θ_{IQ}
ε	<i>errorSignal</i>	the demodulated PDH error signal (Eq. (33))
ζ	<i>computePdh2OmegaSignalDotProduct</i>	the demodulated $2\omega_m$ signal (Eq. (34))

Table 5: Notation used throughout Section 3.

Each spectral line $\ell \in \{-1, 0, 1\}$ carries the cavity's own reflection coefficient, evaluated at that line's own round-trip phase,

$$\varphi_\ell \equiv \frac{2\pi(\Delta f_{\text{detune}} + \ell f_m)}{\text{FSR}}, \quad (23)$$

the same $\varphi = 2\pi\Delta f_{\text{detune}}/\text{FSR}$ of Eq. (3) (§1.3), just evaluated at $\Delta f_{\text{detune}} + \ell f_m$ instead of Δf_{detune} alone — $\varphi_0 = \varphi$. Applying $r(\varphi_\ell)$ to line ℓ 's own phasor amplitude (§2.4's $\mathcal{E}_\ell$) gives that line's own reflected phasor,

$$\mathcal{E}_{\text{refl},\ell} \equiv \mathcal{E}_\ell r(\varphi_\ell), \quad \ell \in \{-1, 0, 1\}, \quad (24)$$

so that recombining on the photodiode, the reflected field's baseband envelope is

$$E_{\text{refl}}(t) = \sum_{\ell=-1}^1 \mathcal{E}_{\text{refl},\ell} e^{i\ell\omega_m t}, \quad (25)$$

Eq. (10) with each line's own phasor now carried through the cavity — exactly the “each frequency component picks up its own value of the cavity's reflection coefficient” of §2.6, written out. A photodiode measures power, the squared magnitude:

$$P_{\text{PD}}(t) = |E_{\text{refl}}(t)|^2 = \sum_{\ell=-1}^1 \sum_{\ell'=-1}^1 \mathcal{E}_{\text{refl},\ell} \mathcal{E}_{\text{refl},\ell'}^* e^{i(\ell-\ell')\omega_m t}. \quad (26)$$

Grouping Eq. (26)'s nine terms by $k = \ell - \ell' \in \{-2, -1, 0, 1, 2\}$ sorts them by frequency: the three $k = 0$ terms carry no time dependence at all, the four $k = \pm 1$ terms (two distinct (ℓ, ℓ') pairs each giving $k = +1$ and $k = -1$)

oscillate at ω_m , and the two $k = \pm 2$ terms oscillate at $2\omega_m$. Writing the full split with each group its own named function of time,

$$P_{\text{PD}}(t) = P_{\text{PD,DC}} + P_{\text{PD},\omega_m}(t) + P_{\text{PD},2\omega_m}(t), \quad (27)$$

we take each term in turn.

The DC term. The three $k = 0$ terms ($\ell = \ell'$) sum to

$$P_{\text{PD,DC}} = \sum_{\ell=-1}^1 |\mathcal{E}_{\text{refl},\ell}|^2, \quad (28)$$

the incoherent sum of each line's own reflected power, obtained directly without simulating any time domain at all (*dcReflectedPower* in *simulatePdhErrorSignal.m* computes exactly this, bypassing Eqs. (25)–(26) entirely).

The ω_m term. The four $k = \pm 1$ terms are the two pairs $(\ell, \ell') = (1, 0)$ and $(0, -1)$:

$$\mathcal{E}_{\text{refl},1} \mathcal{E}_{\text{refl},0}^* e^{i\omega_m t} + \mathcal{E}_{\text{refl},0} \mathcal{E}_{\text{refl},-1}^* e^{i\omega_m t}.$$

Unlike the raw $J_\ell(\beta)$ products, these two terms need no further sign-flip identity invoked here: $\mathcal{E}_{-1} = -\mathcal{E}_1$ (§2.4, itself inherited from $J_{-1}(\beta) = -J_1(\beta)$) is already carried inside $\mathcal{E}_{\text{refl},-1}$, so the whole $e^{i\omega_m t}$ coefficient is simply their sum, the beat-note coefficient

$$C \equiv \mathcal{E}_{\text{refl},1} \mathcal{E}_{\text{refl},0}^* + \mathcal{E}_{\text{refl},0} \mathcal{E}_{\text{refl},-1}^*, \quad (29)$$

giving

$$P_{\text{PD},\omega_m}(t) \equiv C e^{i\omega_m t} + C^* e^{-i\omega_m t} = 2 \text{Re}[C e^{i\omega_m t}], \quad (30)$$

real as a physical power's piece must be, since the $k = +1/k = -1$ pair are each other's complex conjugates (and, by the same token, so are the two $k = \pm 2$ terms making up $P_{\text{PD},2\omega_m}(t)$ below). This is the term §3.3's mixer and low-pass filter extract into the PDH error signal ε .

The $2\omega_m$ term. The two $k = \pm 2$ terms are the pair $(\ell, \ell') = (1, -1)$:

$$\mathcal{E}_{\text{refl},1} \mathcal{E}_{\text{refl},-1}^* e^{i2\omega_m t},$$

already a single term — the beat-note coefficient here is simply

$$D \equiv \mathcal{E}_{\text{refl},1} \mathcal{E}_{\text{refl},-1}^*, \quad (31)$$

giving

$$P_{\text{PD},2\omega_m}(t) \equiv D e^{i2\omega_m t} + D^* e^{-i2\omega_m t} = 2 \text{Re}[D e^{i2\omega_m t}], \quad (32)$$

real as a physical power's piece must be, since the two $k = \pm 2$ terms are each other's complex conjugates. This is the term §3.3's doubled-LO mixer and its own low-pass filter extract into the demodulated $2\omega_m$ signal ζ . Unlike C , D never involves φ_0 at all — it depends only on the two sidebands' own reflection coefficients, a fact Figure 12 below puts to direct use.

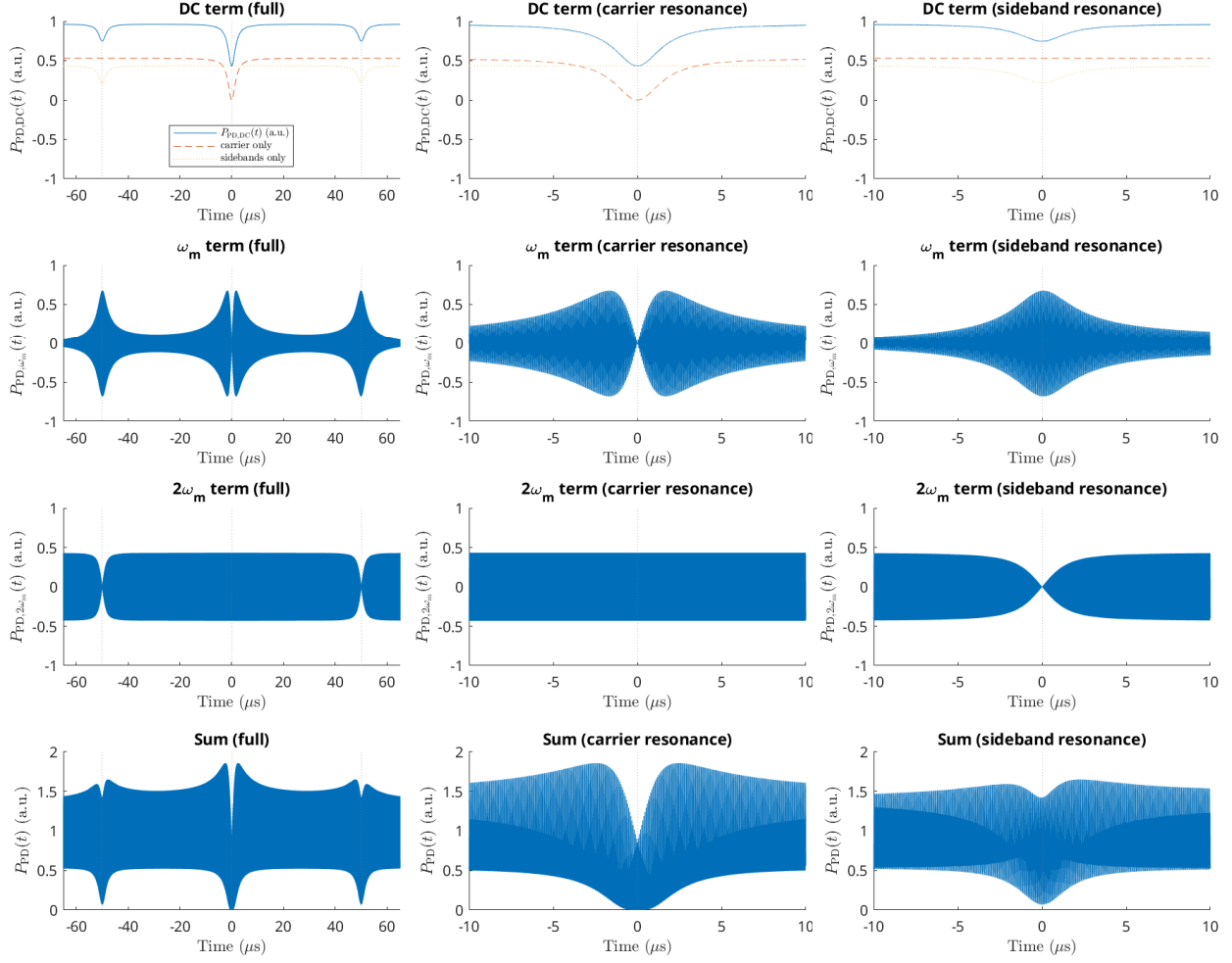


Figure 12: $P_{PD,DC}(t)$, $P_{PD,\omega_m}(t)$, $P_{PD,2\omega_m}(t)$, and their sum, over three matched windows (Figure 13 shares them), all on $[-1, 1]$ except the sum (point 5): a full $\pm 1.3f_m$ overview crossing all three resonances, and a closer time-domain view (± 3 linewidths) around the carrier resonance and the representative sideband resonance at $\Delta f_{\text{detune}} = -f_m$ ($+f_m$ mirrors it) — at this width the fast oscillation is no longer individually resolved, only its envelope. Time is referenced to $t = 0$ at each column's own resonance; dotted lines mark every resonance visible within a column. (1) $P_{PD,DC}(t)$ dips but never reaches zero: only the carrier's own reflection nulls at critical coupling (dashed “carrier only”); the sidebands, f_m away, keep reflecting almost fully (dotted “sidebands only”), setting the dip's floor. (2) In the carrier-resonance column, $P_{PD,\omega_m}(t)$'s envelope pinches to a node and re-emerges inverted right at resonance — the zero-crossing Figure 13 shows already demodulated into ε . (3) $P_{PD,2\omega_m}(t)$'s envelope is essentially constant there, since D (Eq. (31)) depends only on the sidebands' reflection coefficients, barely moved by this scan. (4) The sideband-resonance column breaks that constancy: that sideband's own field nulls, killing D completely, while C only loses one summand and survives — $P_{PD,\omega_m}(t)$ shows a same-signed bump instead of a clean sign-flip; the carrier resonance alone kills the ω_m term. (5) The sum exceeds 1 even though each line individually reflects and transmits without loss (§1.3): only the *time-averaged* reflected+transmitted power matches the time-averaged input (both below 1, §2.2) — the cavity's frequency-dependent response redistributes power *in time* between the coherent lines, like dispersion reshaping a pulse, not energy creation.

3.3 Demodulation

For the PDH signal, the goal is to isolate the *amplitude* of $P_{\text{PD}}(t)$'s ω_m component, $P_{\text{PD},\omega_m}(t)$ (Eq. (30)), while preserving its *phase* — that sign is exactly what tells a servo which side of resonance the laser sits on. The way to do this is downmixing: multiply $P_{\text{PD}}(t)$ by a reference signal at ω_m (the LO) and recover the amplitude by low-pass filtering the result down to its own DC component.

The mixer. The mixer multiplies $P_{\text{PD}}(t)$ by the LO,

$$\cos(\omega_m t + \theta_{\text{demod}}) = \frac{1}{2} \left[e^{i(\omega_m t + \theta_{\text{demod}})} + e^{-i(\omega_m t + \theta_{\text{demod}})} \right],$$

so every resulting cross-term sits at 0, $\pm\omega_m$, or $\pm 2\omega_m$ — never in between — and the low-pass filter below keeps only the first: the pair built from C , the ω_m beat-note coefficient already derived in §3.2 (Eq. (29)).

The same trick extracts the $2\omega_m$ terms too, one harmonic up: mixing $P_{\text{PD}}(t)$ against a frequency-doubled LO,

$$\cos(2\omega_m t + 2\theta_{\text{demod}}) = \frac{1}{2} \left[e^{i(2\omega_m t + 2\theta_{\text{demod}})} + e^{-i(2\omega_m t + 2\theta_{\text{demod}})} \right],$$

isolates D 's $k = \pm 2$ terms (Eq. (31)), making up $P_{\text{PD},2\omega_m}(t)$, exactly as the ω_m LO isolated C 's above. The reference phase doubles along with the frequency because a real frequency doubler derives this second LO from the very same oscillator driving the ω_m mixer: squaring $\cos(\omega_m t + \theta_{\text{demod}})$ produces a $2\omega_m$ term with phase $2\theta_{\text{demod}}$, not a second, independently adjustable phase. This second channel — D , the same coefficient defined in §3.2 (Eq. (31)) — is carried through its own low-pass filter alongside ε below, the same way.

The low-pass filter. Exactly two of the mixer's cross-terms land at zero frequency: pairing $C e^{i\omega_m t}$ with the LO's $e^{-i(\omega_m t + \theta_{\text{demod}})}$ half gives $\frac{1}{2} C e^{-i\theta_{\text{demod}}}$, and pairing $C^* e^{-i\omega_m t}$ with $e^{i(\omega_m t + \theta_{\text{demod}})}$ gives its own complex conjugate, $\frac{1}{2} C^* e^{i\theta_{\text{demod}}}$. Every other cross-term — $P_{\text{PD},\text{DC}} \cos(\omega_m t + \theta_{\text{demod}})$ itself, and the $k = \pm 2$ terms against the LO — sits at $\pm\omega_m$ or $\pm 2\omega_m$ and is removed by the low-pass filter (*simulatePdhErrorSignal.m* designs a 4th-order IIR filter at one-tenth of f_m , applied zero-phase via *filtfilt*). What survives is real by construction, a sum of conjugates:

$$\varepsilon(\Delta f_{\text{detune}}) \equiv \frac{1}{2} \left[C e^{-i\theta_{\text{demod}}} + C^* e^{i\theta_{\text{demod}}} \right] = \text{Re} \left[C e^{-i\theta_{\text{demod}}} \right], \quad (33)$$

the PDH error signal — exactly what *simulatePdhErrorSignal.m* returns, verified there against this same closed form to within numerical noise, cross-checked independently of its own time-domain simulation.

The doubled-LO channel above survives the same way. Pairing $D e^{i2\omega_m t}$ with that LO's $e^{-i(2\omega_m t + 2\theta_{\text{demod}})}$ half gives $\frac{1}{2} D e^{-i2\theta_{\text{demod}}}$, its conjugate pairing gives $\frac{1}{2} D^* e^{i2\theta_{\text{demod}}}$, and every other cross-term is removed by that channel's own low-pass filter, leaving

$$\zeta(\Delta f_{\text{detune}}) \equiv \frac{1}{2} \left[D e^{-i2\theta_{\text{demod}}} + D^* e^{i2\theta_{\text{demod}}} \right] = \text{Re} \left[D e^{-i2\theta_{\text{demod}}} \right], \quad (34)$$

the demodulated $2\omega_m$ signal — *computePdh2OmegaSignalDotProduct.m* computes exactly this closed form directly; no time-domain simulation of this second channel exists in this project, mirroring the closed-form shortcut §3.4 previews for ε itself. Unlike ε , which is odd in Δf_{detune} (Eq. (33)'s S-curve), ζ is *even*: D pairs $r(\Delta f_{\text{detune}} + f_m)$ with $r^*(\Delta f_{\text{detune}} - f_m)$, and $r(-\varphi) = r^*(\varphi)$ (immediate from Eq. (7), since r_1, r_2 are both real) makes that product symmetric under $\Delta f_{\text{detune}} \rightarrow -\Delta f_{\text{detune}}$. ζ also vanishes exactly at each sideband resonance ($\Delta f_{\text{detune}} = \mp f_m$, where that sideband's own field nulls and kills D outright, §3.2) but — unlike ε — does *not* null at the carrier resonance ($\Delta f_{\text{detune}} = 0$), since D never involves φ_0 at all.

Eq. (33) also shows concretely why the demodulation phase in §2.1's “mixing down” step is a real design choice, not a formality: θ_{demod} rotates C before taking its real part, and C 's own phase need not point the same way at every detuning. At $\Delta f_{\text{detune}} = 0.5 \Delta \nu_{\text{FWHM}}$ (this project's running example, $\beta \approx 1.08$), $\varepsilon = 0.339$ at $\theta_{\text{demod}} = \pi/2$ but only $\varepsilon = 0.0113$ at $\theta_{\text{demod}} = 0$ — almost $30\times$ smaller, not merely scaled down — while at

$\Delta f_{\text{detune}} = f_m + 0.5 \Delta \nu_{\text{FWHM}}$ (the secondary, sideband-resonance feature Figure 13's sideband-resonance column shows), the two phases swap roles: $\theta_{\text{demod}} = 0$ gives the larger signal there. The carrier-locking feature this section exists to produce and the sideband-resonance feature live in orthogonal demodulation quadratures, which is why this project's default is $\theta_{\text{demod}} = \pi/2$, not 0.

Figure 13 revisits the exact three windows Figure 12 used in the time domain, now already demodulated: $P_{\text{PD,DC}}$, ε , and ζ over the identical detuning ranges, so the two figures can be read side by side without their windows ever silently drifting apart. The carrier-resonance column is the same clean, sign-flipping zero-crossing Figure 12 showed as a raw envelope pinch in $P_{\text{PD},\omega_m}(t)$; the sideband-resonance column is the smaller, same-signed bump that same term showed there too, now easy to read off directly as a curve. ζ 's own row makes Eq. (34)'s even symmetry and its exact sideband-resonance null immediately visible, mirroring $P_{\text{PD},2\omega_m}(t)$'s envelope in the time domain.

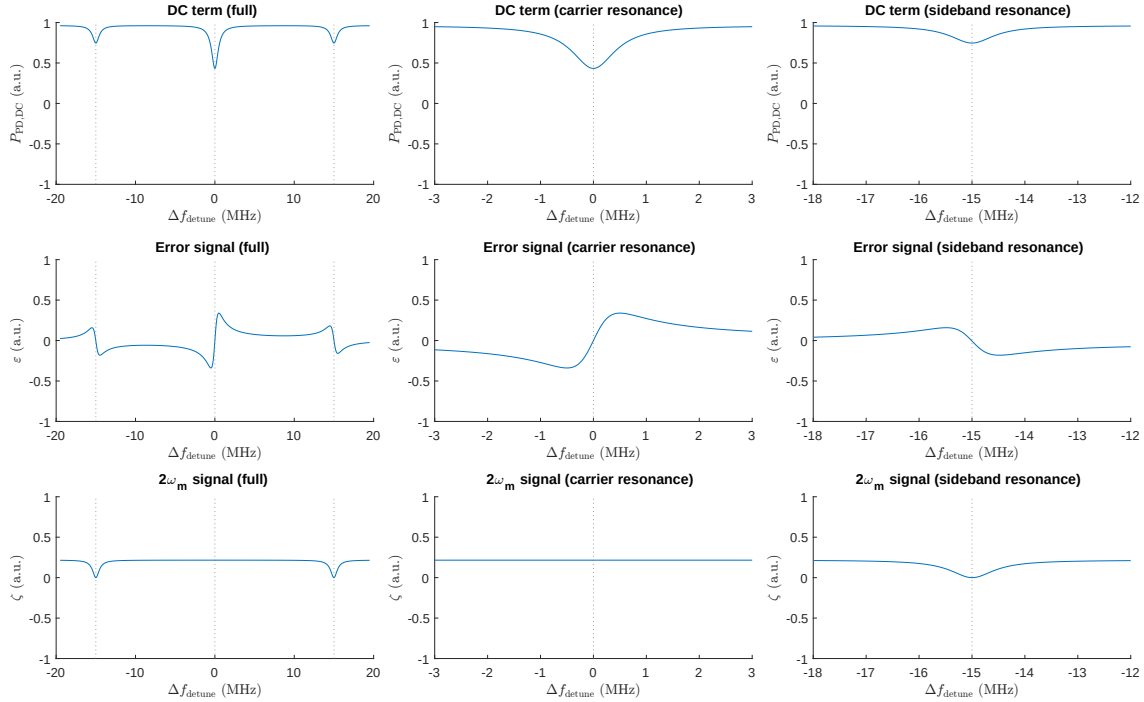


Figure 13: $P_{\text{PD,DC}}$, the demodulated error signal ε , and its $2\omega_m$ counterpart ζ (Eq. (34)) vs. detuning, over the exact same three windows (full, carrier resonance, sideband resonance) as Figure 12's time-domain view, all on the same $[-1, 1]$ vertical scale — both figures read the same underlying spans (*pdhPhotodiodeScanSpans.m*), so they cannot drift out of sync with each other. The carrier-resonance column is the dispersive S-curve of Eq. (33) — ζ shows no feature there at all, since D never involves φ_0 ; the sideband-resonance column is ε 's secondary feature from the θ_{demod} discussion above, together with ζ 's exact null there.

3.4 Where this leads

Eq. (33) was reached by simulating a photodiode, a mixer, and a filter numerically — exactly what *simulatePdhErrorSignal.m* does, sample by sample, at every detuning. None of that simulation was actually necessary: C (Eq. (29)) is built entirely from the three lines' own reflected fields, with no time dependence left in it at all, and $\varepsilon = \text{Re}(Ce^{-i\theta_{\text{demod}}})$ is nothing more than a fixed rotation of a fixed complex number. Section 4 makes that explicit: it shows C is a phasor dot product between the carrier's own reflected field and a combination of the two sidebands' — the same ε , reached without ever constructing $E_{\text{refl}}(t)$, $P_{\text{PD}}(t)$, or a single time sample.

4 The Phasor Dot-Product Shortcut

4.1 The dot-product identity

Table 6 collects the one symbol new to this section; everything else ($\mathcal{E}_\ell$, $\mathcal{E}_{\text{refl},\ell}$, θ_{demod} , C , ε) reuses Sections 2 and 3 exactly as already defined.

Symbol	Code identifier	Meaning
$\mathcal{E}_{\text{refl},\text{sb}}$	—	the sideband resultant phasor, the two sidebands' own reflected phasors combined at the demodulation phase (Eq. (37))

Table 6: Notation used throughout Section 4.

For any two complex numbers z, w , treating each as a real 2D vector (Re, Im) makes an otherwise-opaque piece of complex algebra literal:

$$\text{Re}(zw^*) = \text{Re}(z) \text{Re}(w) + \text{Im}(z) \text{Im}(w) \equiv z \cdot w = |z| |w| \cos(\phi_z - \phi_w), \quad (35)$$

the ordinary real dot product of the two vectors — exactly *PhasorMath.dotProduct*, and, via the cosine form (writing $z = |z|e^{i\phi_z}$, $w = |w|e^{i\phi_w}$), the length of w 's projection onto z 's own direction, scaled by $|z|$: it is maximized when the two phasors point the same way ($\phi_z = \phi_w$), zero when they are perpendicular, and flips sign once they point more than 90° apart. Section 3.2 already built the beat-note coefficient $C = \mathcal{E}_{\text{refl},1}\mathcal{E}_{\text{refl},0}^* + \mathcal{E}_{\text{refl},0}\mathcal{E}_{\text{refl},-1}^*$ (Eq. (29)) and the error signal $\varepsilon = \text{Re}[Ce^{-i\theta_{\text{demod}}}]$ (Eq. (33)) out of it — a sum in which the carrier's own reflected phasor $\mathcal{E}_{\text{refl},0}$ appears, once conjugated and once not, multiplying each sideband's reflected phasor in turn. Substituting C into ε and using $\text{Re}(u^*) = \text{Re}(u)$ on the $\mathcal{E}_{\text{refl},1}$ term (to flip which factor carries the conjugate without changing the real part it contributes) lets the shared factor $\mathcal{E}_{\text{refl},0}$ be pulled out from under a single, common conjugate:

$$\varepsilon = \text{Re}\left[\mathcal{E}_{\text{refl},0} \left(\mathcal{E}_{\text{refl},1}e^{-i\theta_{\text{demod}}} + \mathcal{E}_{\text{refl},-1}e^{i\theta_{\text{demod}}}\right)^*\right]. \quad (36)$$

Naming the parenthesized sum

$$\mathcal{E}_{\text{refl},\text{sb}} \equiv \mathcal{E}_{\text{refl},1}e^{-i\theta_{\text{demod}}} + \mathcal{E}_{\text{refl},-1}e^{i\theta_{\text{demod}}} \quad (37)$$

— the two sidebands' own reflected phasors, each rotated by the demodulation phase in opposite directions (since they counter-rotate, §2.4) and summed — turns Eq. (36) into exactly Eq. (35)'s dot product:

$$\varepsilon = \text{Re}[\mathcal{E}_{\text{refl},0} \mathcal{E}_{\text{refl},\text{sb}}^*] = \mathcal{E}_{\text{refl},0} \cdot \mathcal{E}_{\text{refl},\text{sb}}. \quad (38)$$

This is the promise §3.4 closed on: the same error signal already derived in §3.3, understood geometrically, is the dot product of the carrier's own reflected phasor with the (demodulation-phase-combined) sideband resultant — two fixed complex numbers at any given detuning.

What the demodulation phase actually controls. $\mathcal{E}_{\text{refl},1}e^{-i\theta_{\text{demod}}}$ and $\mathcal{E}_{\text{refl},-1}e^{i\theta_{\text{demod}}}$ each rotate rigidly, oppositely, as θ_{demod} sweeps — but their sum $\mathcal{E}_{\text{refl},\text{sb}}$ generally does not: two counter-rotating phasors of unequal magnitude trace an ellipse, not a circle, as θ_{demod} completes a turn. That ellipse degenerates to a rigid rotation only where one term vanishes outright — §4.2's side-mode-resonance case is exactly this: the upper term is zero there, so $\mathcal{E}_{\text{refl},\text{sb}}(\theta_{\text{demod}}) = \mathcal{E}_{\text{refl},-1}e^{i\theta_{\text{demod}}}$ alone, and it does rotate rigidly, verified to floating-point precision. Away from such a point the resultant's own path is generally more complicated than either a rotation or a fixed direction, so nothing about I and Q being orthogonal can be read off its motion in general. What *is* always true, with no such caveat, is a direct consequence of Eq. (33): $\varepsilon(\theta_{\text{demod}}) = \text{Re}[Ce^{-i\theta_{\text{demod}}}]$ is a pure cosine in θ_{demod} , of amplitude $|C|$ and phase offset $\arg C$, regardless of how $\mathcal{E}_{\text{refl},\text{sb}}$ itself moves — Eq. (38) identifies this cosine

as $\mathcal{E}_{\text{ref},0}$ dotted against $\mathcal{E}_{\text{ref},\text{sb}}(\theta_{\text{demod}})$. Reading $\theta_{\text{demod}} = 0$ and $\theta_{\text{demod}} = \pi/2$ off that one cosine gives exactly $\text{Re}(C)$ and $\text{Im}(C)$: Q and I are the real and imaginary parts of the single complex number this section's dot product already computes, which is why they are orthogonal — in the ordinary sense that real and imaginary parts are — without either sideband resultant needing to visibly rotate through 90° to make them so. This is exactly *computePdhErrorSignalDotProduct.m* (*PhasorMath.dotProduct* applied to *PhasorMath.sidebandAmplitude* and *PhasorMath.combineAtDemodulationPhase*'s outputs), already cross-checked against §3.3's time-domain ε to within numerical noise.

4.2 A detuning-by-detuning walkthrough

A single representative snapshot of Eq. (38)'s phasor picture is not enough to see *why* the dot product takes the values it does as detuning actually sweeps through the cavity's resonance structure. Figure 14 below evaluates the PDH cavity at four characteristic detunings, one row each, showing all four faces of the picture together: (a) the phasor diagram — carrier phasor (green), the two sideband phasors combined into the sideband resultant (blue/red arrows and their black sum), and the dot product between them (shaded projection) — fixed to a common $[-1, 1]$ scale so magnitude changes between rows are directly comparable rather than hidden by auto-scaling; (b) the resulting PDH error signal $\varepsilon(\Delta f)$ of Eq. (38), marked at that row's own detuning; (c) the reflected power $|r(\Delta f)|^2$; and (d) the reflected phase $\arg r(\Delta f)$, unwrapped — the same construction *animatePdhDetuningScanPhasors.m*'s live tool draws, frozen at four illustrative points instead of animated. All four rows share one background detuning span (the same $1.3f_m$ span the live tool itself uses), and panels (c)/(d) mark all three lines on that shared curve at once: the carrier's own detuning (green), the upper sideband's (blue), and the lower sideband's (red) — three evaluations of the identical response function, not three different curves.

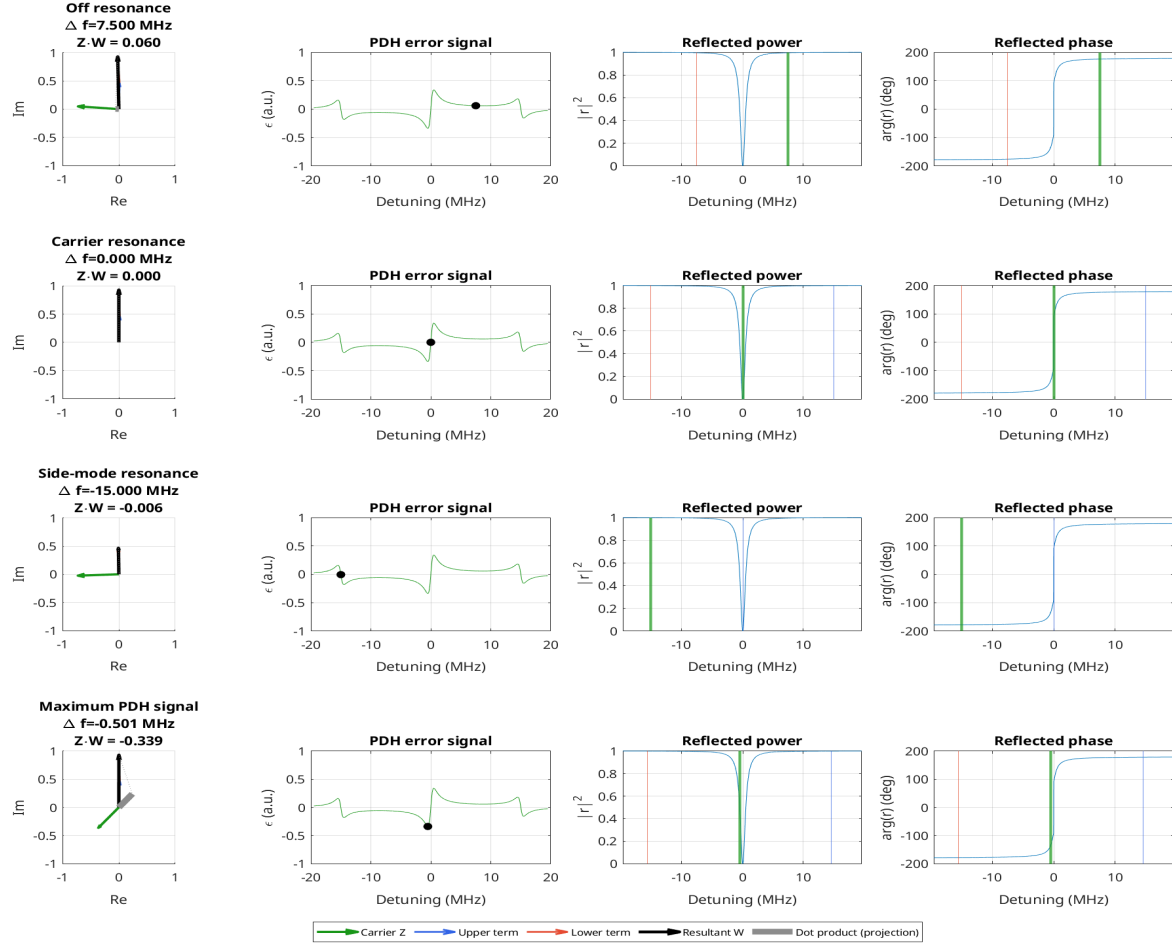


Figure 14: Four characteristic detunings of the PDH cavity example, one row each: off resonance, carrier resonance, side-mode resonance, and the detuning that numerically maximizes $|\varepsilon(\Delta f)|$. Column (a) phasors on a common $[-1, 1]$ scale; (b) the error signal with a marker at the row's detuning; (c)/(d) reflected power/phase with markers at the carrier's (green), upper sideband's (blue), and lower sideband's (red) own detuning.

Case 1: off resonance ($\Delta f = f_m/2$). This detuning is 7.5 linewidths from the carrier's own resonance and, by construction, equally 7.5 linewidths from each sideband resonance at $\pm f_m$ — the most off-resonance point available inside the shared span. All three lines therefore sit far enough into the background that their reflectivity is nearly identical: $|r(\Delta f)|^2 = 0.996$, close to the mirror's own asymptotic background, with $\arg r(\Delta f) = 176.2^\circ$, just short of the 180° a truly infinite detuning would give — f_m is only 15 linewidths from the carrier's resonance, not infinitely

many, so a small residual survives here deliberately rather than being rounded away. Because all three lines share nearly the same phase, $\mathcal{E}_{\text{refl},0} \approx -0.725 + 0.048i$ sits almost exactly on the negative real axis, and so does the raw upper-sideband term $\mathcal{E}_{\text{refl},1} \approx -0.465 + 0.010i$, while the raw lower term is very nearly its negative, $\mathcal{E}_{\text{refl},-1} \approx -\mathcal{E}_{\text{refl},1}$, from $\mathcal{E}_{-1} = -\mathcal{E}_1$ (§2.2) acting on two nearly-equal reflectivities. With $\mathcal{E}_{\text{refl},-1} = -\mathcal{E}_{\text{refl},1}$, Eq. (37) collapses to

$$\mathcal{E}_{\text{refl},\text{sb}}(\theta_{\text{demod}}) = \mathcal{E}_{\text{refl},1} (e^{-i\theta_{\text{demod}}} - e^{i\theta_{\text{demod}}}) = -2i \mathcal{E}_{\text{refl},1} \sin \theta_{\text{demod}}$$

— $\mathcal{E}_{\text{refl},1}$ rotated by exactly 90° , for every θ_{demod} and every modulation depth β (exact to floating-point precision with no cavity at all). The 90° angle is fixed by phase modulation’s own sideband symmetry, not by any tunable parameter of this system. The one thing that moves it away from 90° is **detuning**: close enough to an actual resonance, the cavity stops giving the carrier and the sidebands the same phase, and $\arg \mathcal{E}_{\text{refl},1} \neq \arg \mathcal{E}_{\text{refl},0}$ — exactly the 84.9° , not 90° , measured here. Two (nearly) perpendicular phasors dot to almost nothing: $\varepsilon = 0.060$, about 18% of this cavity’s peak (case 4 below) and the smallest-magnitude of the four cases.

Case 2: carrier resonance ($\Delta f = 0$). The carrier itself now sits exactly on resonance. Critical coupling makes $r(0) = 0$ exactly (§1.4), so $\mathcal{E}_{\text{refl},0}$ vanishes outright — panel (a) shows no carrier arrow at all. The sidebands, still $f_m = 15$ linewidths away, are essentially unmoved from case 1’s own background value ($\mathcal{E}_{\text{refl},\text{sb}} = 0.930i$ here, purely imaginary to three digits). The dot product of anything with the zero vector is zero: $\varepsilon = 0$ exactly, not approximately — a different mechanism from case 1’s near-perpendicularity, even though both land at or near zero.

Case 3: side-mode resonance ($\Delta f = -f_m$). Detuning the carrier by exactly one modulation frequency below its own resonance parks the *upper* sideband exactly on resonance instead ($\Delta f + f_m = 0$), so its own reflected phasor vanishes: the upper term drops out of $\mathcal{E}_{\text{refl},\text{sb}}$ entirely, leaving only the lower term ($\mathcal{E}_{\text{refl},\text{sb}} = -0.008 + 0.466i$, half the magnitude of case 1’s fully-populated resultant). The carrier itself, sitting $f_m = 15$ linewidths from its own resonance, is still essentially at background ($\mathcal{E}_{\text{refl},0} = -0.727 - 0.024i$). Despite one sideband dropping out, the dot product is still small ($\varepsilon = -0.006$): the carrier and the now-smaller resultant are still close to perpendicular, for the same reason as case 1. It is not, however, exactly zero the way case 2 is — this secondary dispersive feature’s true zero-crossing sits a small fraction of a linewidth away from $\Delta f = -f_m$, precisely because $f_m/\Delta\nu_{\text{FWHM}} = 15$ is large but finite, not infinite.

Case 4: maximum PDH signal ($\Delta f = -0.501 \Delta\nu_{\text{FWHM}}$). Found numerically as the detuning that maximizes $|\varepsilon(\Delta f)|$ across the shared span (the same search §3.3 already used to establish this cavity’s own canonical point), this is where the carrier’s own dispersive feature — not any sideband feature — does the work. Here $\mathcal{E}_{\text{refl},0} = -0.365 - 0.364i$: its real and imaginary parts are equal in magnitude to three significant figures, i.e. $\arg(\mathcal{E}_{\text{refl},0}) = -135.1^\circ$, exactly 45° away from case 1’s off-resonance background direction of $\approx 176\text{--}178^\circ$. This is the general extremum condition for a complex, Lorentzian-like response: the component of $\mathcal{E}_{\text{refl},0}$ perpendicular to the background direction — the part a $\theta_{\text{demod}} = \pi/2$ demodulation actually senses — is maximized exactly where the phasor has rotated a further 45° from where it started. The sideband resultant is essentially unchanged from every other case ($\mathcal{E}_{\text{refl},\text{sb}} = 0.001 + 0.930i$, since $\pm f_m$ is unaffected by a sub-linewidth carrier shift), so nearly all of the case-to-case variation in ε traces back to $\mathcal{E}_{\text{refl},0}$ alone: $\varepsilon = -0.339$, $5.7\times$ case 1’s magnitude and this cavity’s largest achievable value.

Reusing the identical four cases with the real cavity response $r(\Delta f)$ replaced everywhere by its magnitude $|r(\Delta f)|$ — an absorption-spectroscopy analog: the dot-product construction does not require a dispersive, phase-only response, so substituting the reflected power’s *magnitude* $|r(\varphi)|$ for $r(\varphi)$ itself (phase discarded, an explicit simplification) turns the same machinery into a model of amplitude-only absorption, evaluated on the same shared detuning axis as Figure 14 — makes the cost of discarding phase concrete. At $\theta_{\text{demod}} = \pi/2$ this substitution is degenerate: $\mathcal{E}_{\text{refl},0}$ is always real (no phase survives $|\cdot|$) while $\mathcal{E}_{\text{refl},\text{sb}}$ is always purely imaginary at that demodulation phase, so their dot product is identically zero at every detuning, to floating-point precision. Figure 15 therefore uses $\theta_{\text{demod}} = 0$ (Q phase) instead, the one demodulation phase that actually senses an amplitude-only feature.

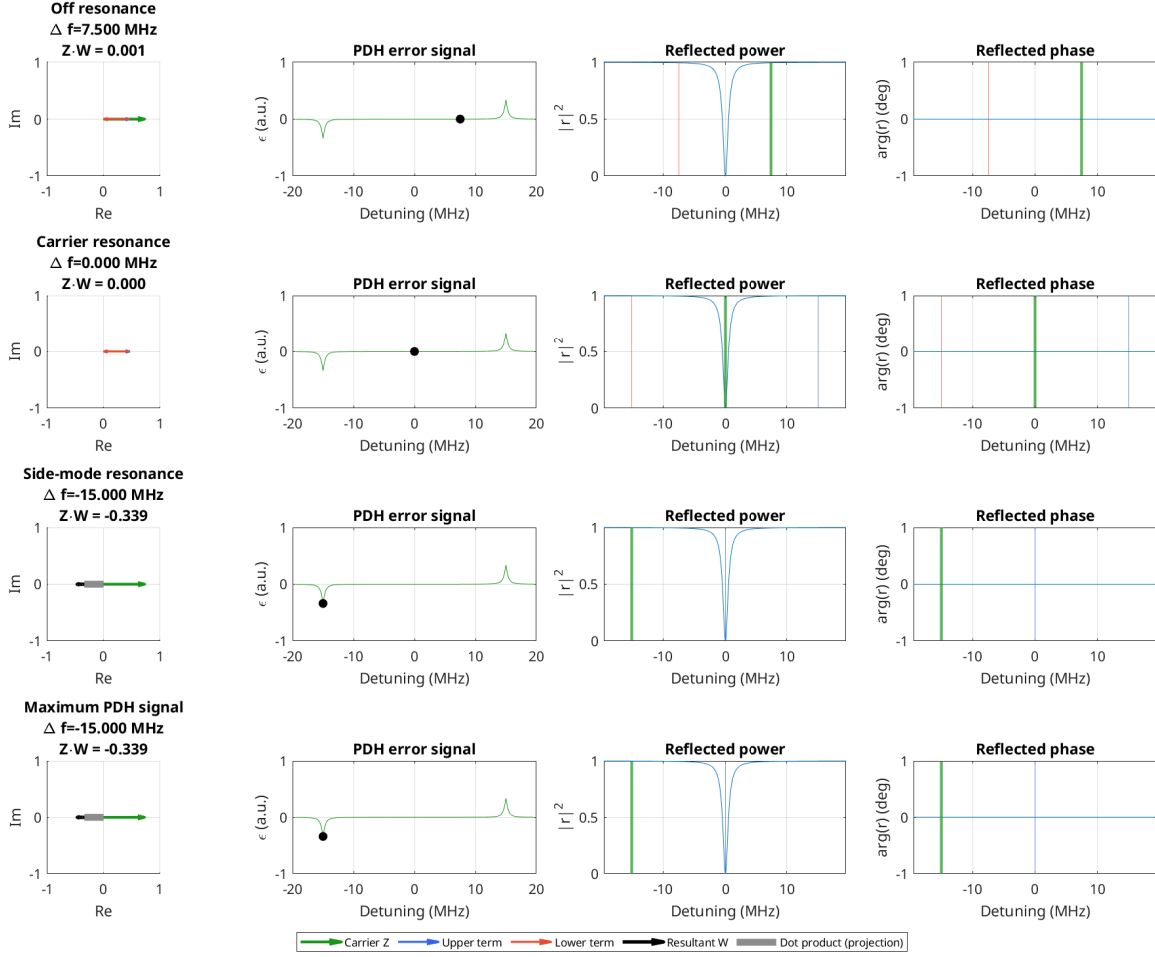


Figure 15: The identical four cases and layout as Figure 14, with the cavity’s complex reflection coefficient replaced by its magnitude everywhere, at $\theta_{\text{demod}} = 0$ (Q phase). Reflected phase is flat at 0° in every row: an amplitude-only detector never sees the phase information Figure 14 relied on. Cases 3 and 4 coincide exactly.

Cases 1 and 2 (absorption-only): still near zero, for a magnitude reason now. Off resonance ($\Delta f = f_m/2$), both sidebands sit at nearly the same background magnitude ($|r(\Delta f + f_m)| \approx |r(\Delta f - f_m)| \approx 0.999$), so their difference — all that survives at $\theta_{\text{demod}} = 0$, since $\mathcal{E}_{-1} = -\mathcal{E}_1$ turns a sum into a difference here — nearly cancels: $\mathcal{E}_{\text{refl, sb}} = 0.0009$, giving $\varepsilon = 0.0007$. On carrier resonance, $\mathcal{E}_{\text{refl, 0}} = 0$ exactly, for the identical critical-coupling reason as case 2 above.

Cases 3 and 4 (absorption-only) coincide. Both land at $\Delta f = -f_m, \varepsilon = -0.339$ — not forced together, found independently (case 3 by definition, case 4 by the same numerical search used in Figure 14) and agreeing to within the search grid’s own resolution, nine orders of magnitude below f_m . The reason is structural, not coincidental: with phase discarded, $\mathcal{E}_{\text{refl, 0}} = \mathcal{E}_0 |r(\Delta f)|$ is real and non-negative, and its magnitude is *minimized*, not extremized, exactly where its slope is steepest ($\Delta f = 0$, critical coupling’s own resonance dip) — the near-resonance tradeoff that produced case 4’s 45° point above (a fast-varying phase carried by a still-substantial magnitude) has no counterpart here, because there is no phase left to carry it. The only remaining source of detuning-dependence is

$\mathcal{E}_{\text{ref},\text{sb}} = \mathcal{E}_1(|r(\Delta f + f_m)| - |r(\Delta f - f_m)|)$, an odd, difference-of-two-dips function of Δf that is itself extremized where one dip is entered fully while the other stays at background — precisely the side-mode-resonance detuning. Losing phase does not merely shrink the PDH error signal: it deletes its near-carrier mechanism outright and replaces it with an on/off sideband-notch effect.

References

- [1] Eric D. Black. An introduction to Pound-Drever-Hall laser frequency stabilization. *American Journal of Physics*, 69(1):79–87, 2001.